Visual Data Analysis
Summer Term 2026

UNIVERSITÄT BONN

Chapter 7: Visualization Design

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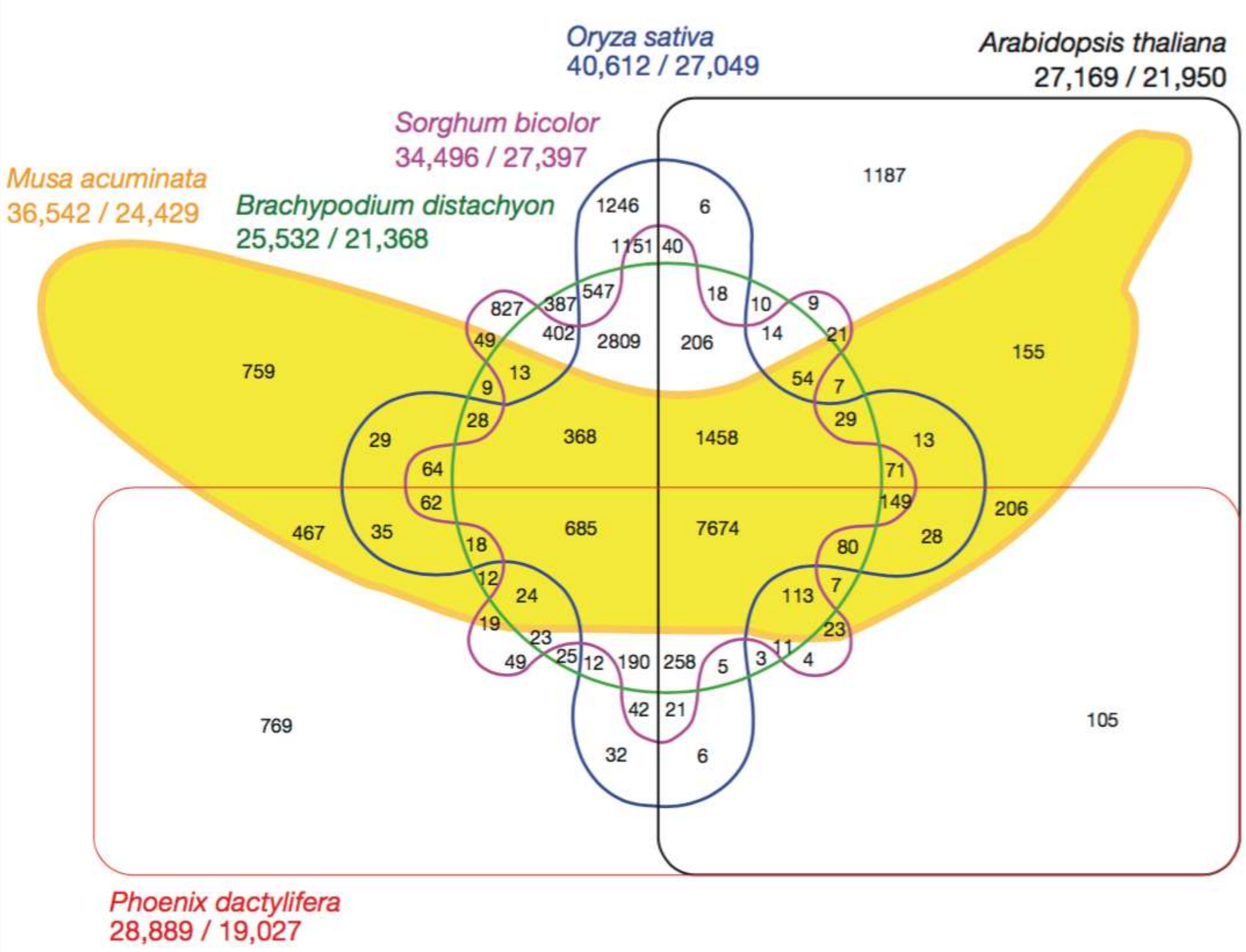
Introduction

- **Knowledge from lecture so far:**
  - Algorithms and techniques:
    - Visualization of multidimensional data
    - Dimensionality Reduction
    - Graph Visualization
    - Geospatial Data Visualization
  - Background on color and perception
- **To come:**
  - Scientific Visualization: Scalar, vector, tensor fields
  - Bonus chapter on neural networks
- **Current chapter:** How to use this knowledge to design and produce effective visualizations?

Section 7.1:
Motivating Examples

Motivating Example 1: Venn Diagram Failure



[D'Hont et al., Nature 2012]

Motivating Example 1: UpSet Visualization

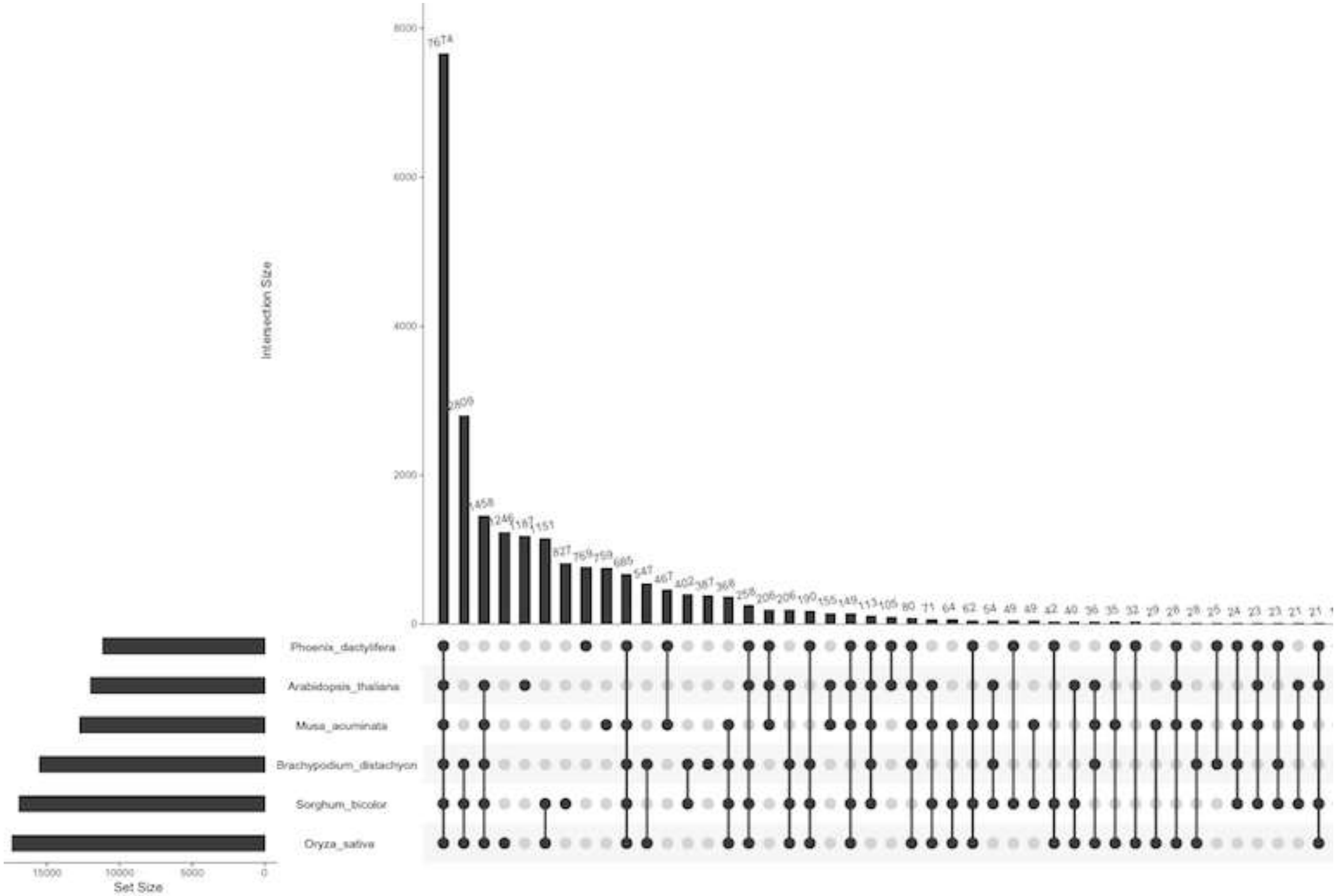


Figure from the Caleydo Team: https://caleydo.org/tools/upset/

Motivating Example 2: Pathline

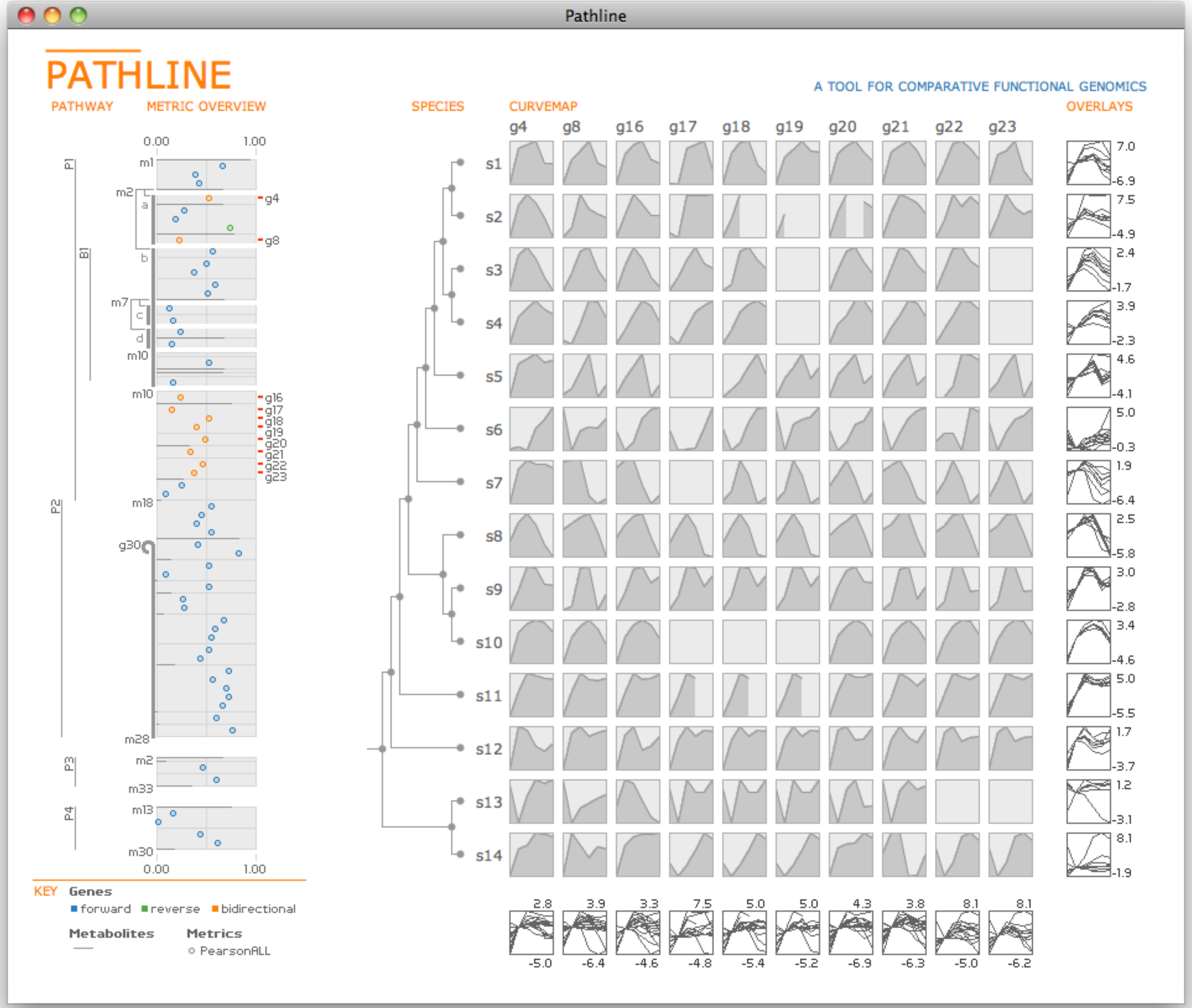
- Design study by [Meyer et al. 2010]
- **Comparative functional genomics**
  – How do gene regulation and metabolite levels in pathways evolve over time and across species?
- **Specific data:** 14 species of yeast
  – **Activity** of 6000 genes and **levels** of 140 metabolites at six points in time
  – **Metabolic pathways:** Directed graph (nodes: metabolites, edges annotated with genes)
  – **Similarity scores:** Similarities between time series measured by Pearson / Spearman correlation
  – **Phylogenetic relationship:** Tree (leaves: species) that shows differentiation of the species

Task Abstraction

- **High-level goal:** Find commonalities and differences between species in the activation patterns of genes and metabolite levels
  – At which point in evolution were specific cellular processes and regulatory mechanisms introduced?
- **More specific tasks:**
  – Look for trends in a set of time series for a gene / metabolite across species
  – Look for trends across genes / metabolites in a set of time series within a species
  – Compare time series to find specific features (e.g., timing of peaks, clusters of time series etc.)

Design: Overview of System

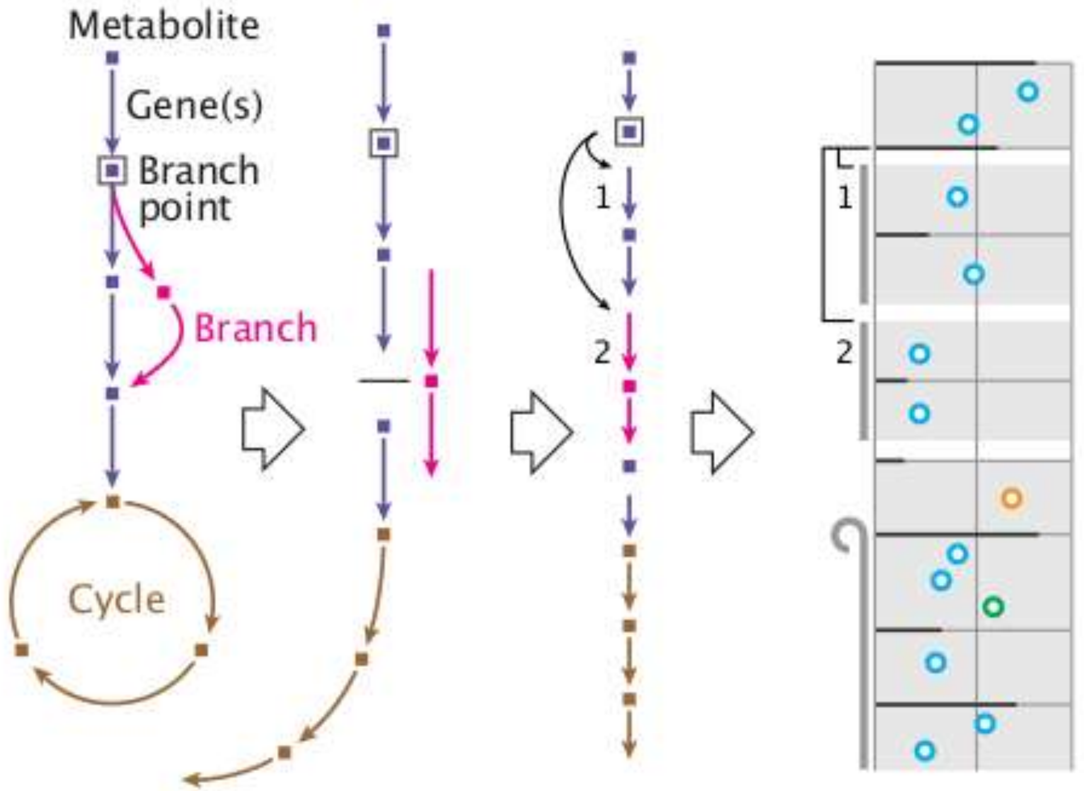
- **Two main views:** Linearized Pathways vs. Curvemap
- Can select genes for analysis in curvemap



Images from [Meyer et al. 2010]

Design: Linearized Pathways

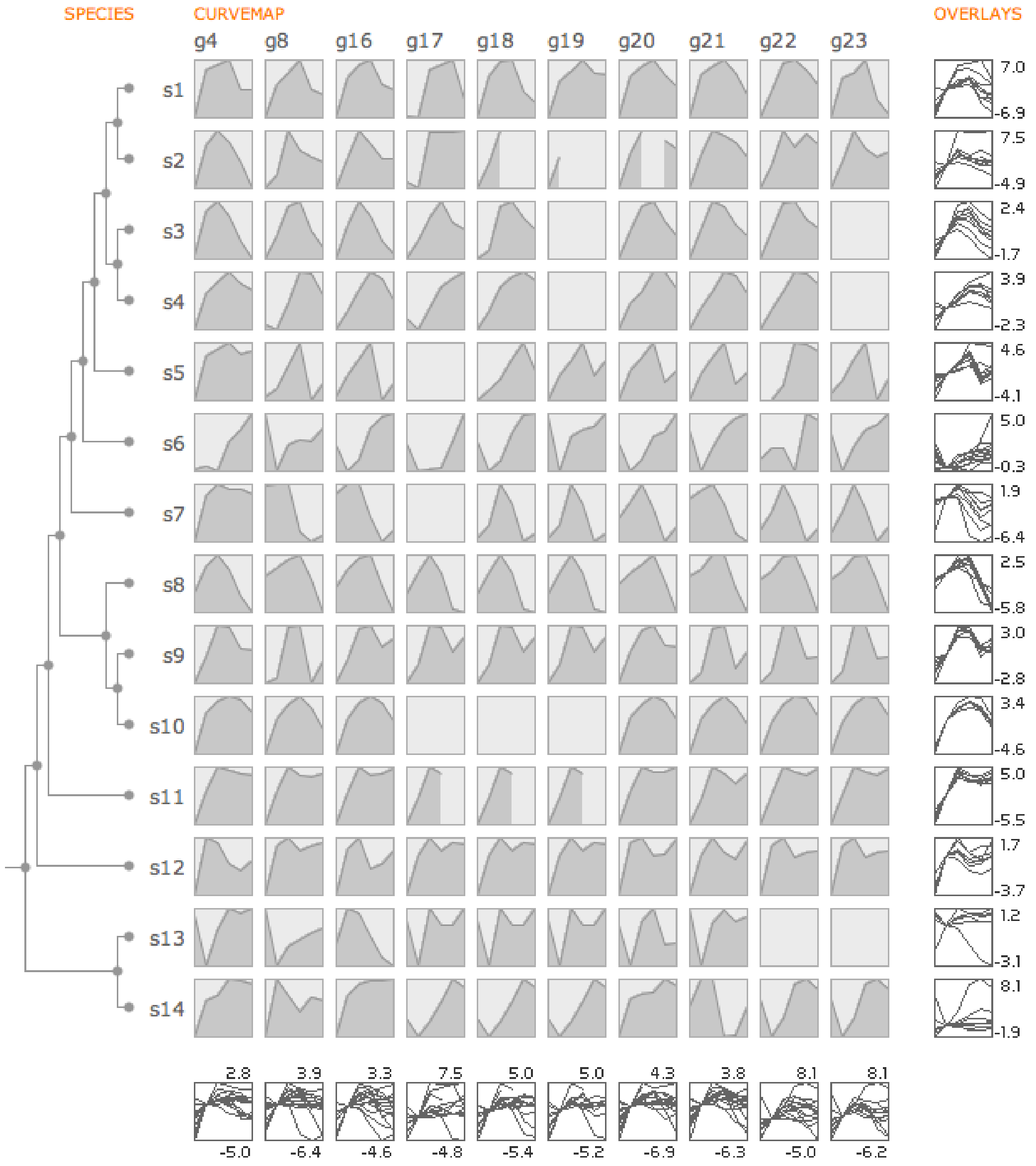
- **Goal:** Emphasize quantitative values along pathways, pathway topology is secondary
  - Unroll loops, disconnect and reinsert branches
  - Gaps indicate branches
  - Only include relevant genes / metabolites
  - Query user in case of ambiguities
  - Lines encode similarity scores of metabolites
  - Circles encode genes
  - Color indicates direction
  - Different pathways shown below each other



Images from [Meyer et al. 2010]

Design: Curvemap View

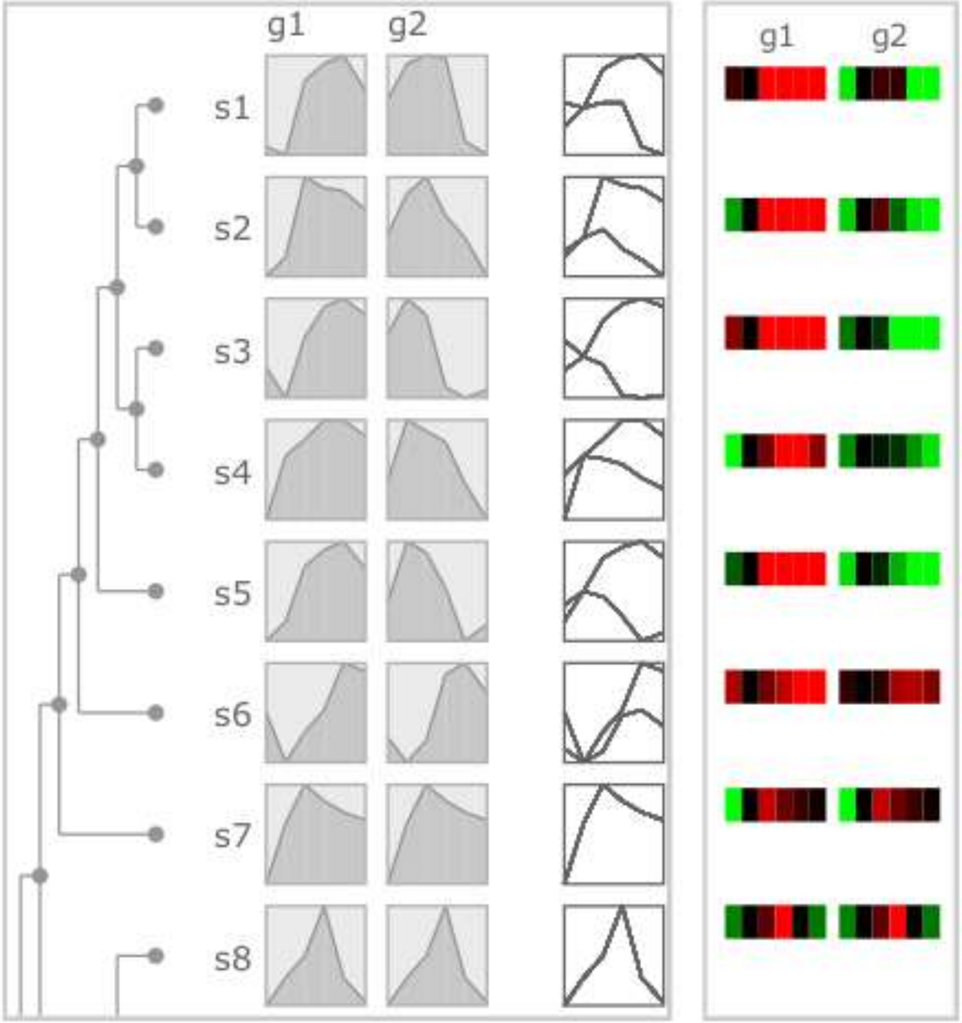
- **Matrix view,** curves in cells show protein expression / metabolite levels over time
  - Better than heat maps
- **Min/max normalization** per cell to make shape obvious
- **Aggregate views** for each row / column for more detailed comparison (incl. absolute levels)
- **Phylogenetic tree** shows how species relate



Images from [Meyer et al. 2010]

Deploy: Case Study 1

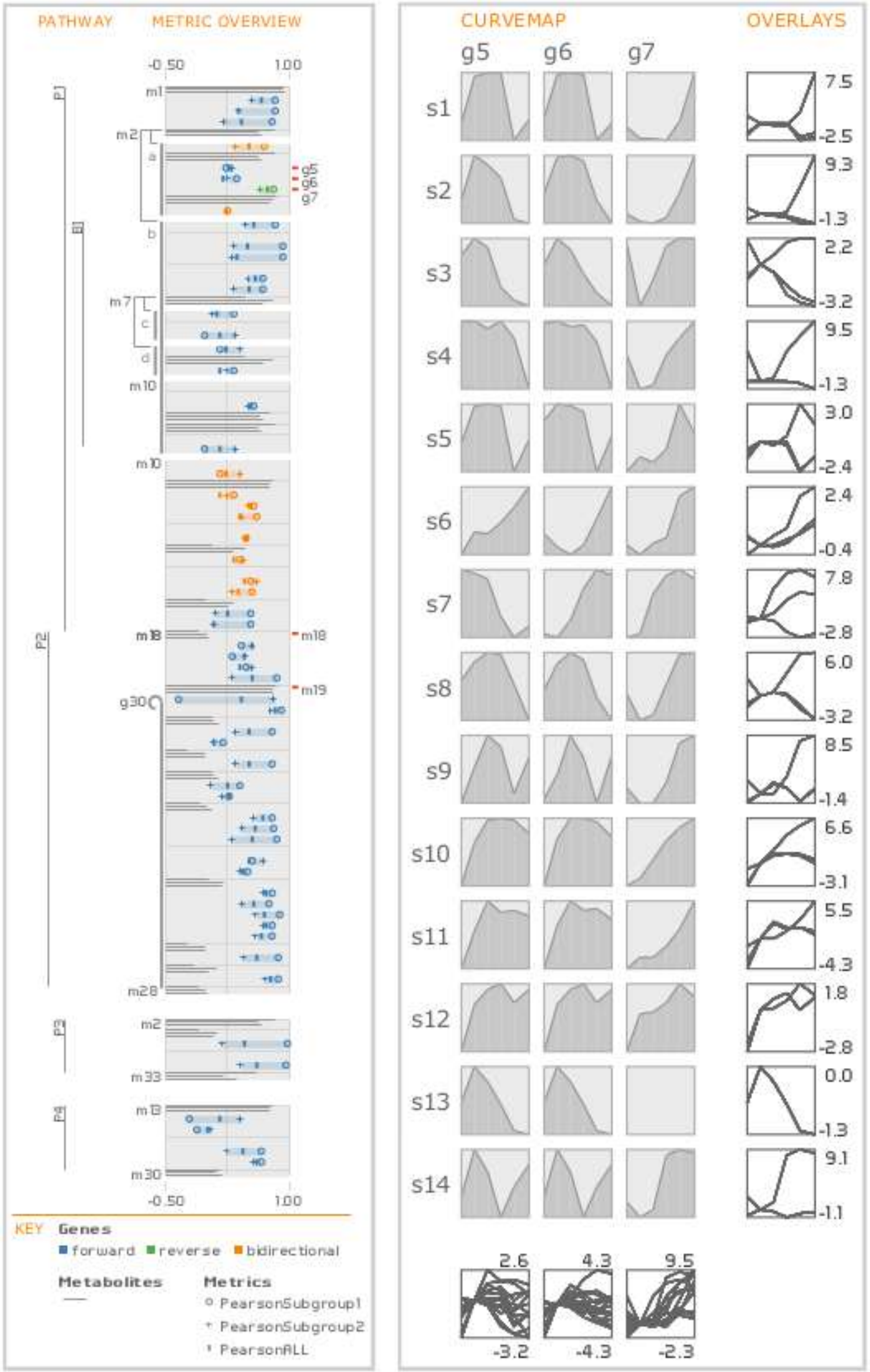
- Investigate **whole-genome duplication**
  - g1/g2 are duplicates in s1-s7
  - Curvemap clearly shows how different regulation mechanisms have evolved
  - Heat maps make it much more difficult to see this



Images from [Meyer et al. 2010]

Deploy: Case Study 2

- **Pathway analysis** (left)
  - Clear trend of reduced similarity in metabolites along the pathway
  - Trend suddenly reversed between m18 and m19
- **Gene-level analysis**
  - g5 and g6 strongly coupled, but time course varies across species
  - Discovered previously unknown gene duplication event in s7



Images from [Meyer et al. 2010]

Summary: Pathline

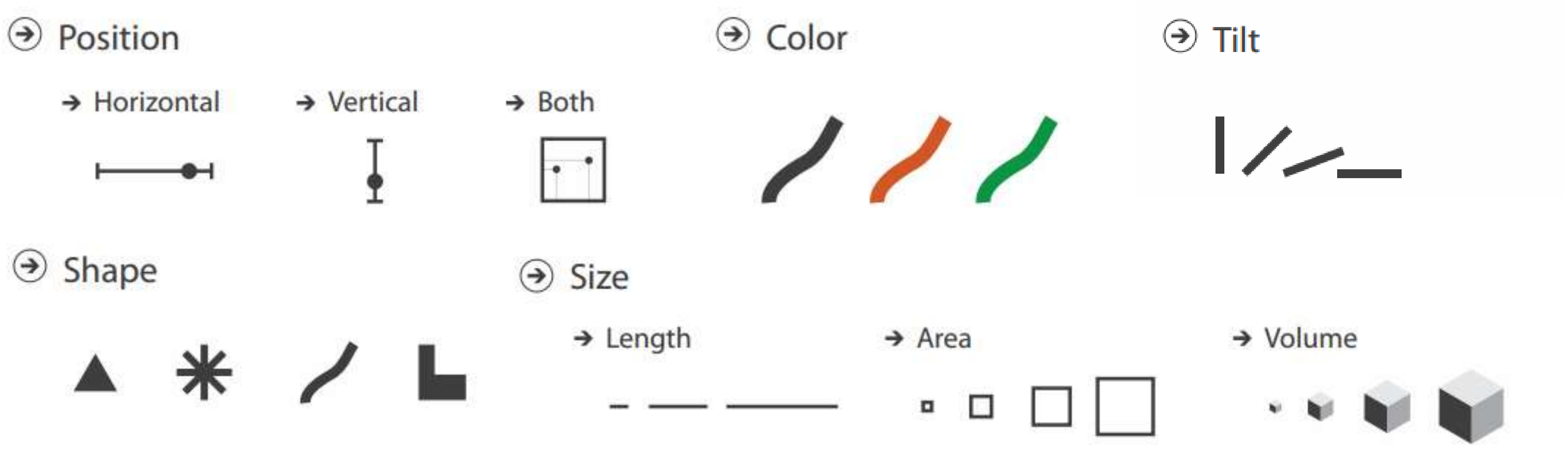
- **Pathline** is an example visualization system that successfully re-designs traditional visual encodings
  - Linearized representation of pathways
  - Curvemap instead of heatmaps
- To guide the design of such systems, we will now discuss
  - Effectiveness of **marks and channels**
  - A **visualization design process** that guides interaction with domain experts, in particular analysis of their **data, tasks, and requirements**
  - An **algebraic perspective** on reasoning about visualization design

Section 7.2:
Marks and Channels

What are Marks and Channels?

- **Marks** are geometric primitives

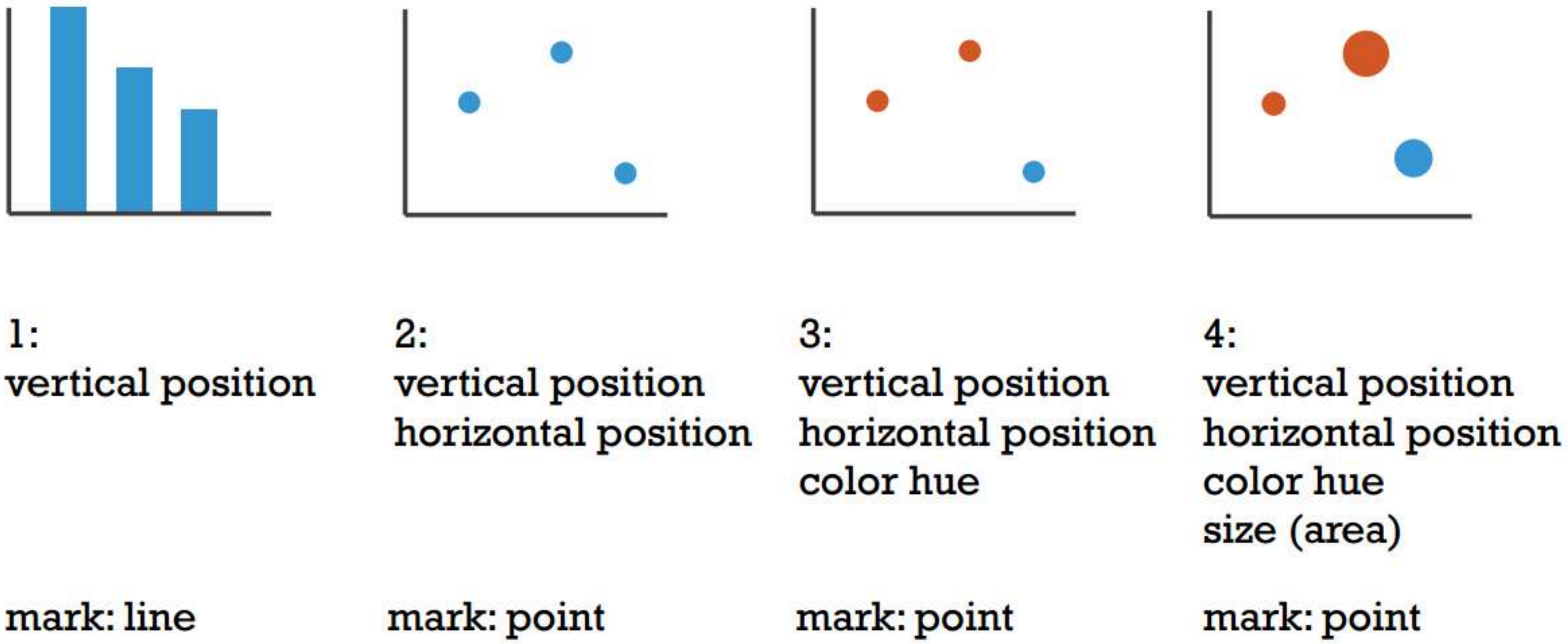
- **Channels** control visual appearance of marks



Images from Tamara Munzner, Visualization Analysis and Design, 2014

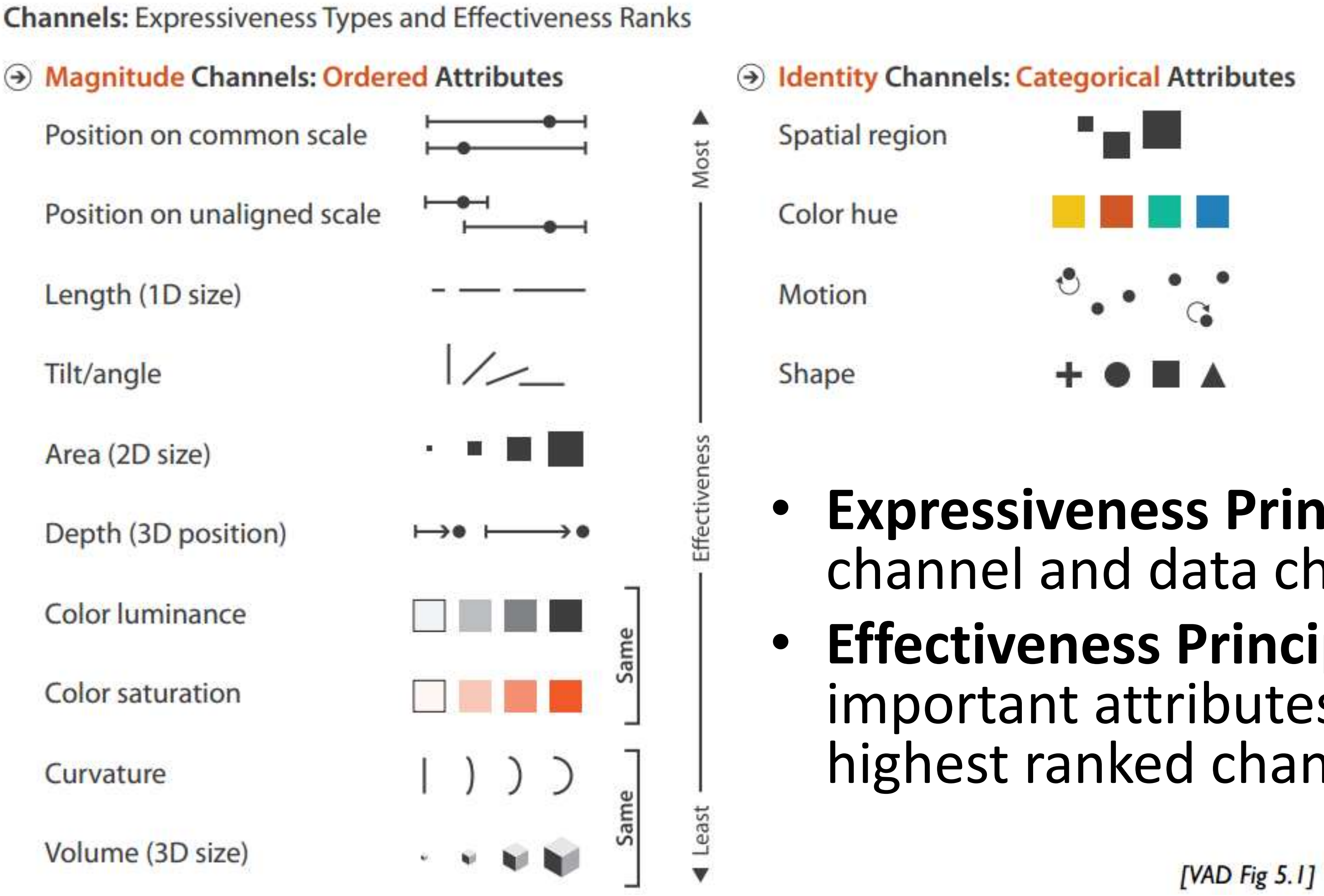
Combining Marks and Channels

- Common visualization idioms can be analyzed as combinations of marks and channels:



Images from Tamara Munzner, Visualization Analysis and Design, 2014

Effectiveness of Channels

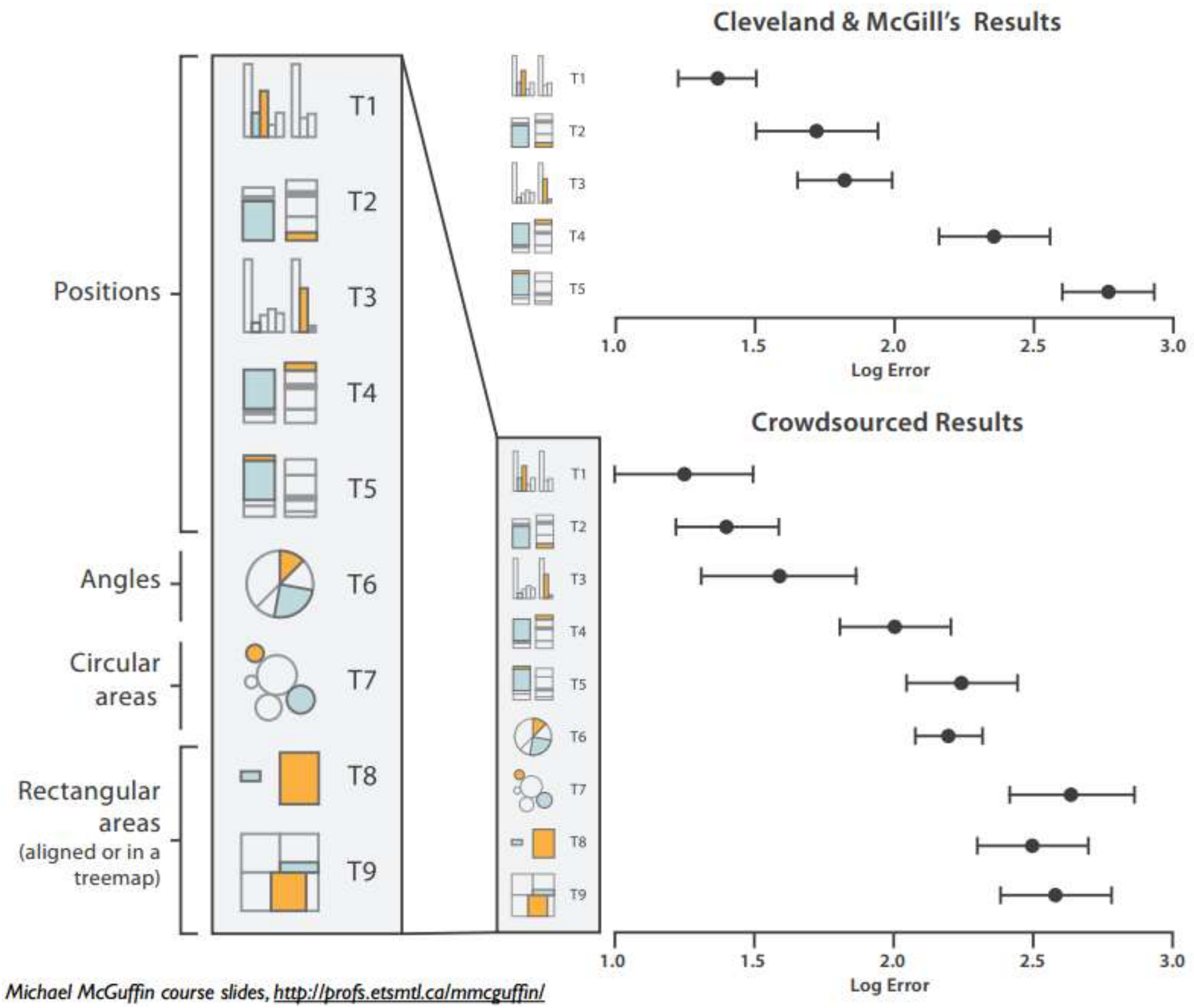


- **Expressiveness Principle:** Match channel and data characteristics
- **Effectiveness Principle:** Most important attributes should use highest ranked channels

Images from Tamara Munzner, Visualization Analysis and Design, 2014

Effectiveness: How do we know?

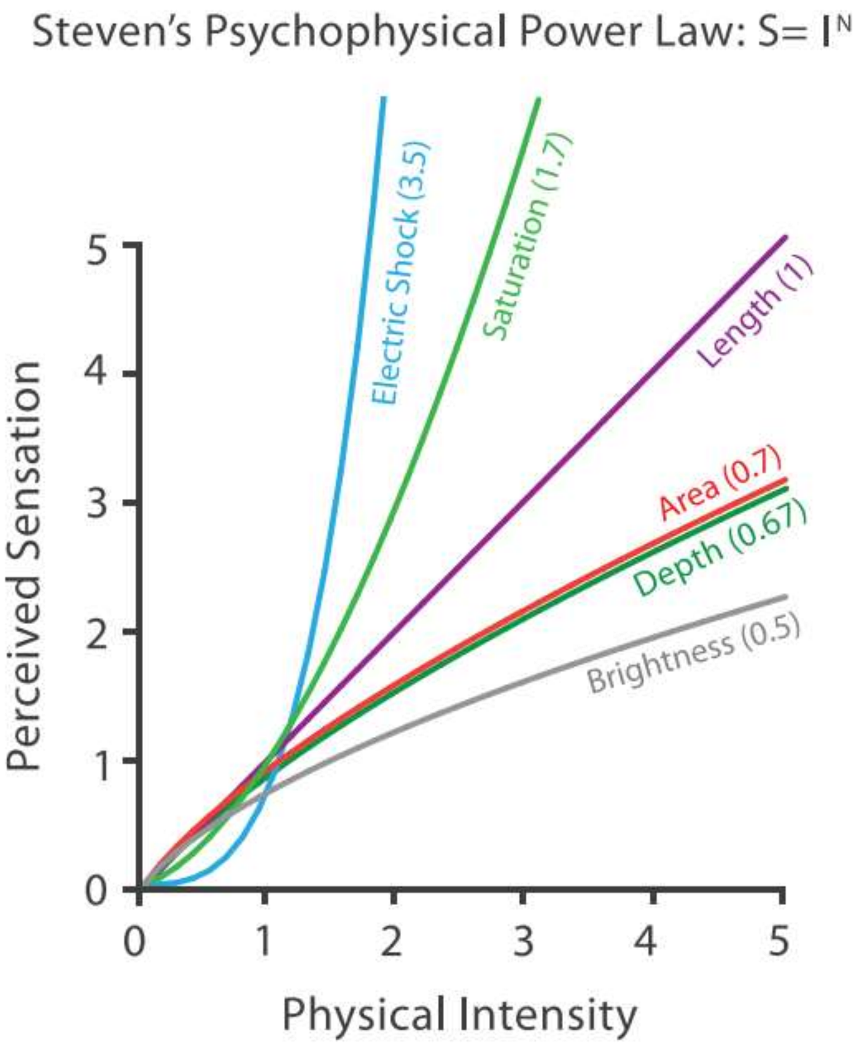
- Compare accuracy (or speed) on clearly defined tasks with different encodings



Michael McGuffin course slides, http://profs.etsmtl.ca/mmcguffin/

Effectiveness: Why?

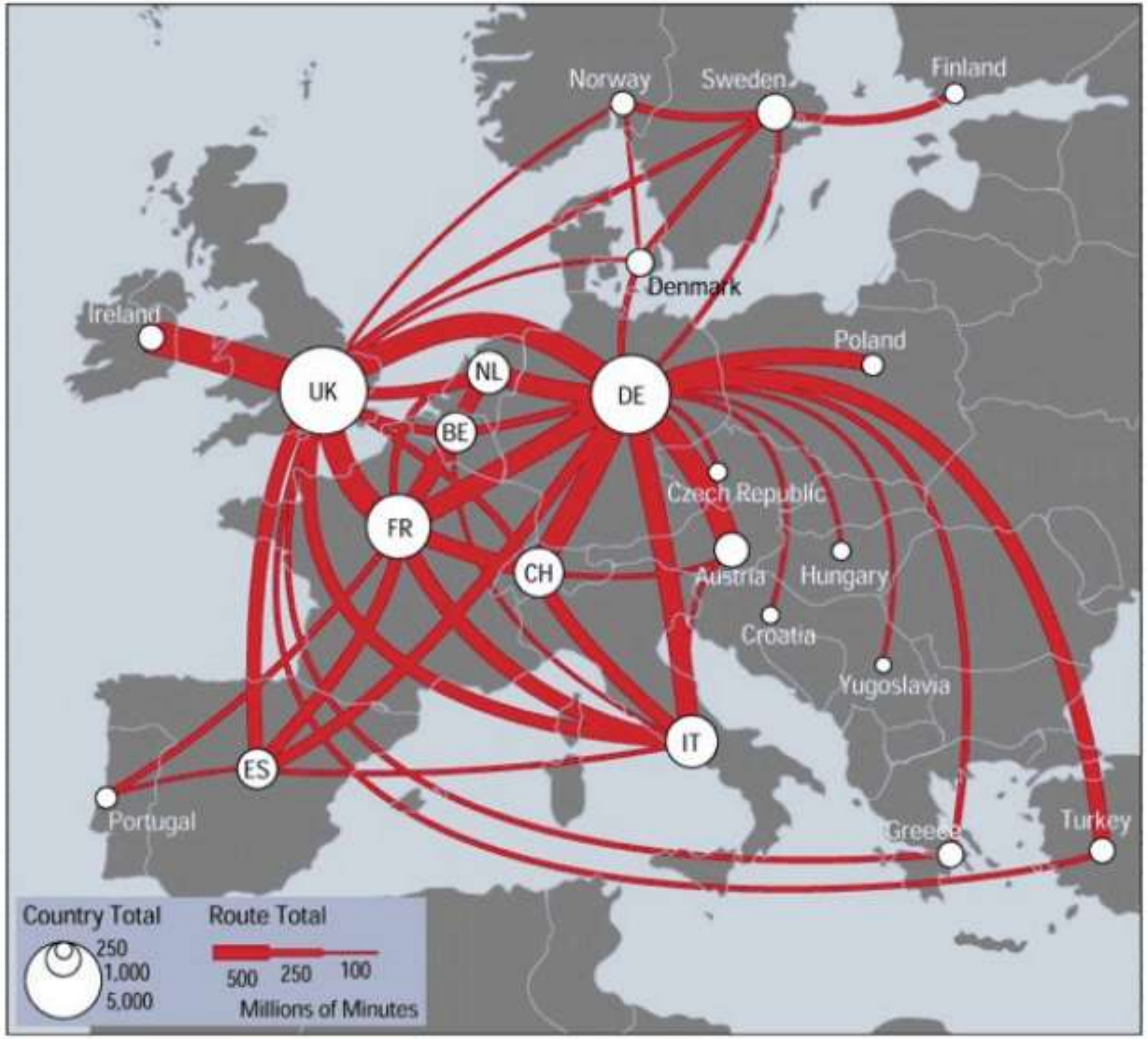
- Differences in accuracy explained by Steven‘s Psychophysical Power Law



Images from Tamara Munzner, Visualization Analysis and Design, 2014

Expressiveness: How Many Levels?

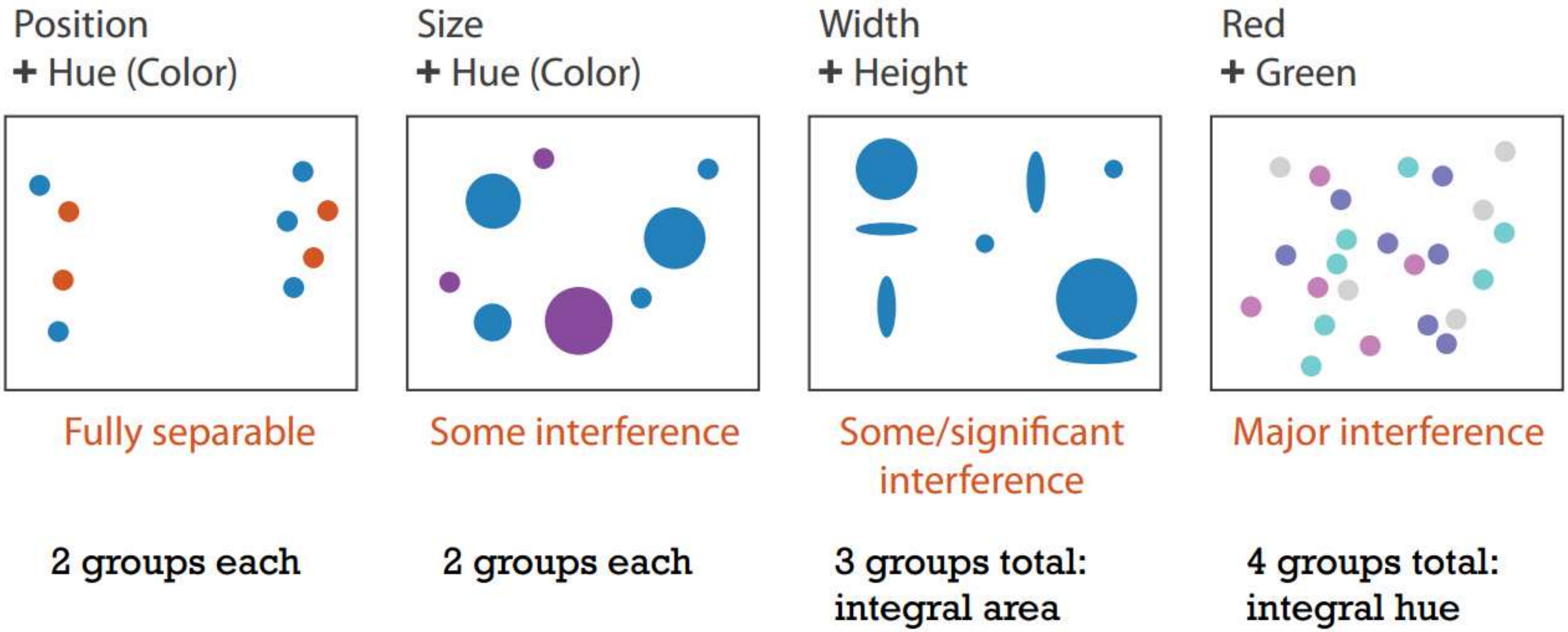
- Some channels only allow us to reliably distinguish between few levels
- Binning in the graph itself and/or the legend can be helpful
  - see discussion in the context of color maps (Section 4.3)



[mappa.mundi.net/maps/maps 014/telegeography.html]

Combinations of Channels

- Keep in mind separability vs. integrability:



Images from Tamara Munzner, Visualization Analysis and Design, 2014

Pre-attentive Processing

- Some channels are processed *preattentively*
  - quickly, without the need for focusing attention
- < 200 - 250ms qualifies as pre-attentive
  - eye movements alone take at least 200ms
  - yet certain processing can be done very quickly, implying low-level processing in parallel
- If a decision takes a fixed amount of time regardless of the number of distractors, it is considered to be preattentive.

Example: Preattentive Processing

- Count the number of 3s:

1281768756138976546984506985604982826762

9809858458224509856458945098450980943585

9091030209905959595772564675050678904567

8845789809821677654876364908560912949686

Example: Preattentive Processing

- Count the number of 3s:

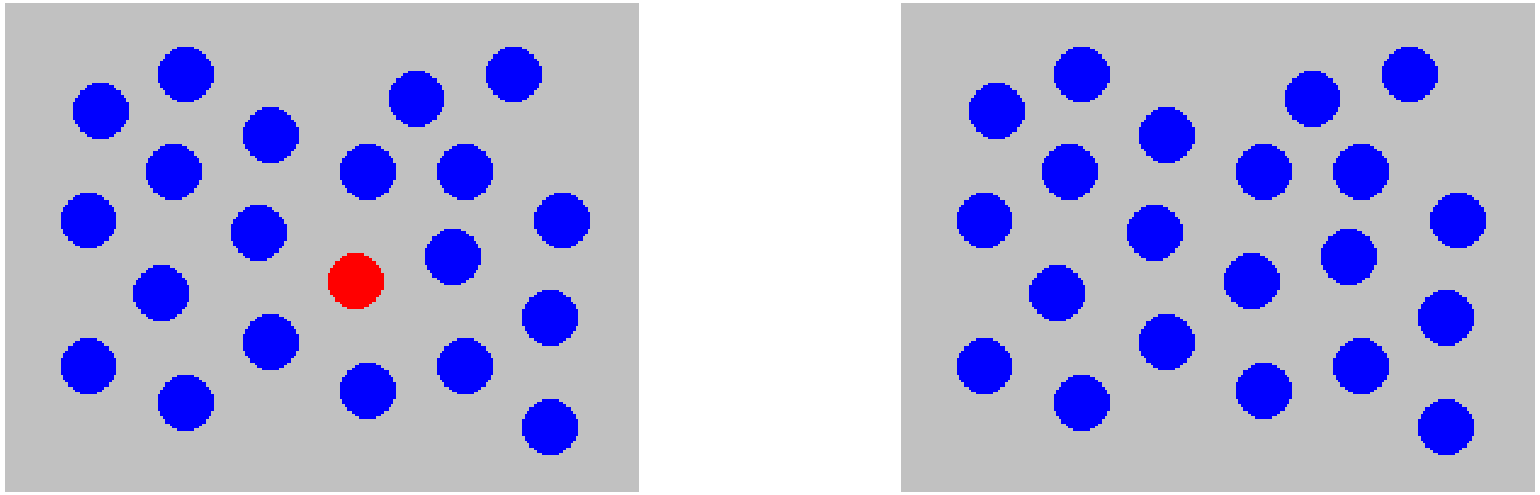
12817687561**3**8976546984506985604982826762

98098584582245098564589450984509809**4**3585

90910**3**0209905959595772564675050678904567

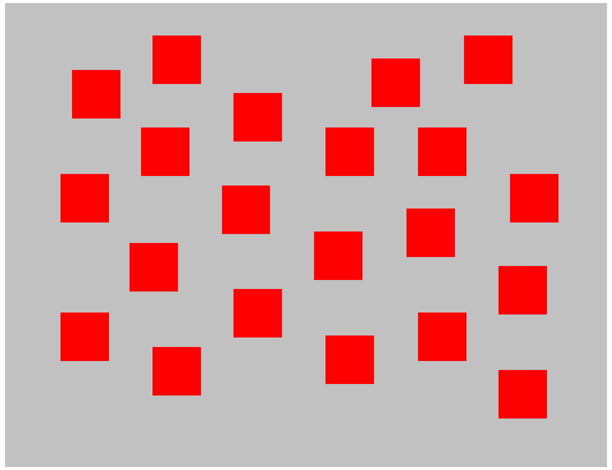
8845789809821677654876**3**64908560912949686

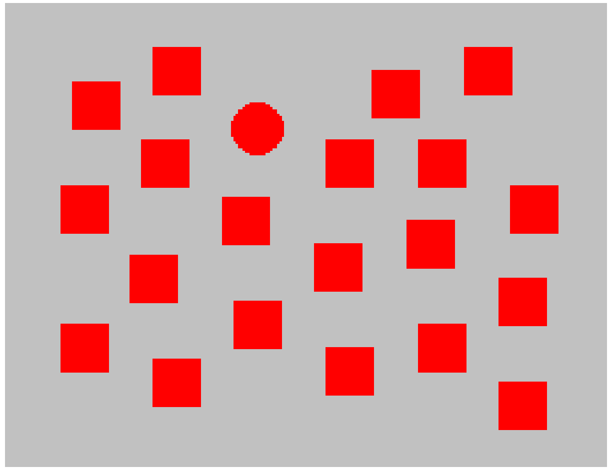
Example: Color Selection



Viewer can rapidly and accurately determine whether the target (red circle) is present or absent. Difference detected in color.

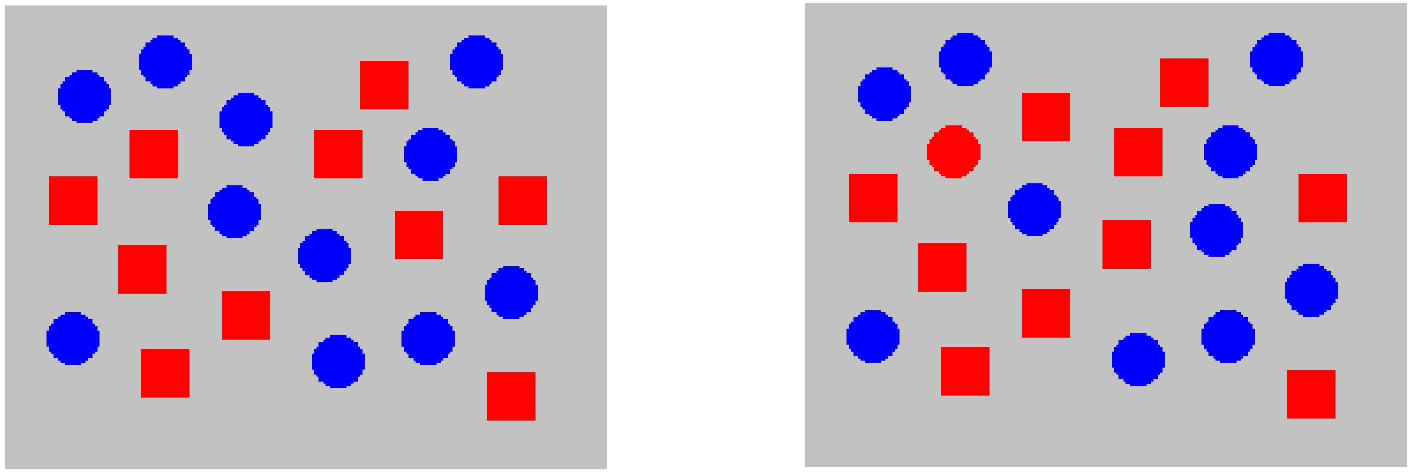
Example: Shape Selection





Viewer can rapidly and accurately determine whether the target (red circle) is present or absent. Difference detected in form (curvature)

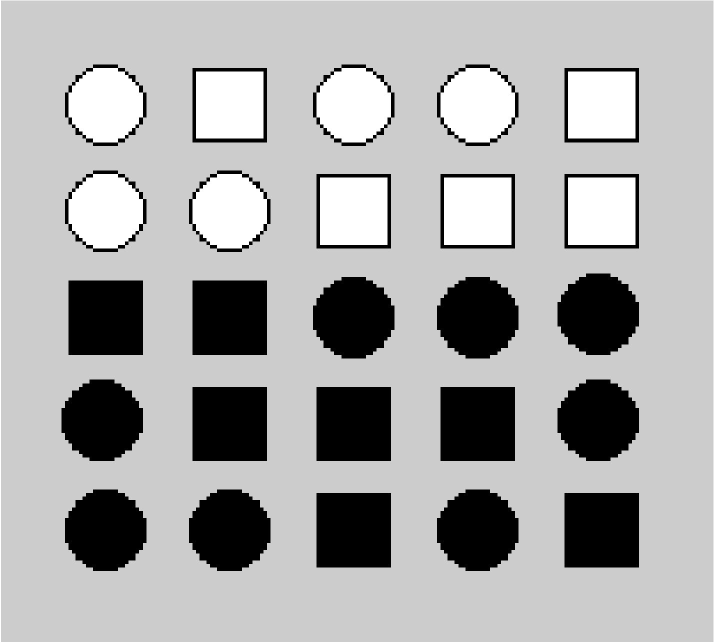
Example: Conjunction of Features

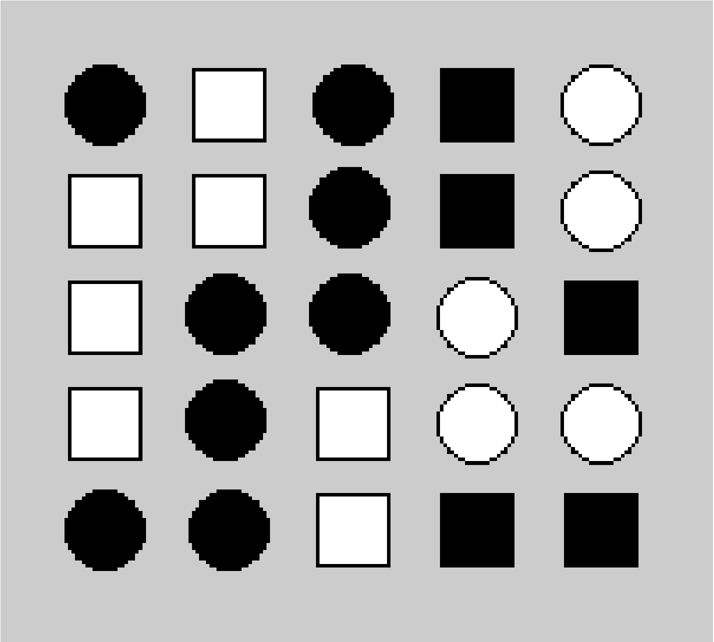


Viewer can *no longer* rapidly and accurately determine whether the target (red circle) is present or absent when target has two or more features, each of which are present in the distractors. Viewer must search sequentially.

Example: Shape vs. Fill

- **Question:** Is there a boundary?

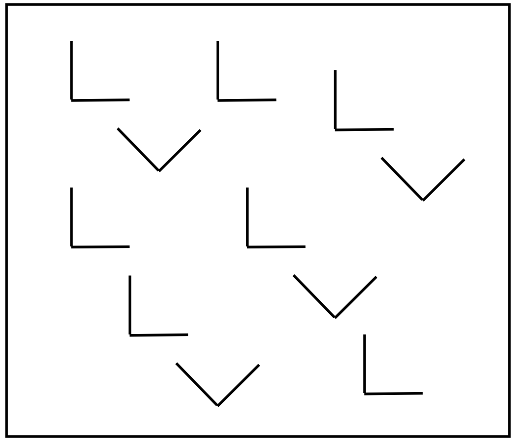


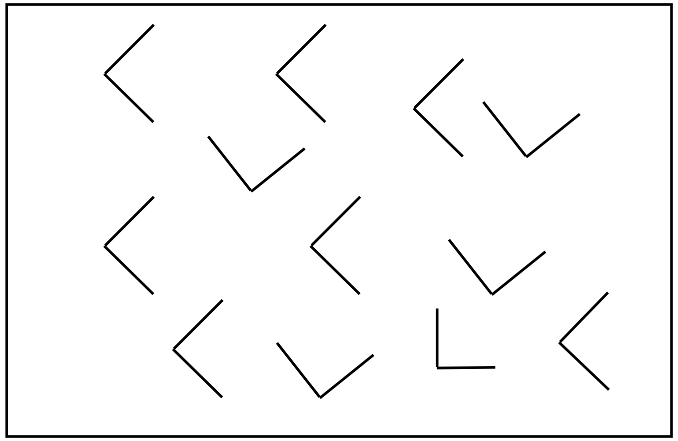


Cannot be done preattentively if both features change at the same time.

Asymmetric Preattentive Properties

- Some properties are asymmetric
  - a sloped line among vertical lines is preattentive
  - a vertical line among sloped ones is not





Text NOT Preattentive

SUBJECT PUNCHED QUICKLY OXIDIZED TCEJBUS DEHCNUP YLKCIUQ DEZIDIXO
CERTAIN QUICKLY PUNCHED METHODS NIATREC YLKCIUQ DEHCNUP SDOHTEM
SCIENCE ENGLISH RECORDS COLUMNS ECNEICS HSILGNE SDROCER SNMULOC
GOVERNS PRECISE EXAMPLE MERCURY SNREVOG ESICERP ELPMAXE YRUCREM
CERTAIN QUICKLY PUNCHED METHODS NIATREC YLKCIUQ DEHCNUP SDOHTEM
GOVERNS PRECISE EXAMPLE MERCURY SNREVOG ESICERP ELPMAXE YRUCREM
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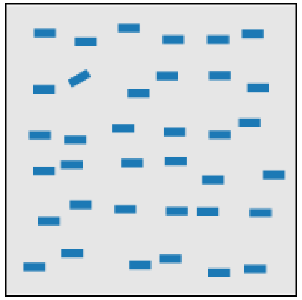
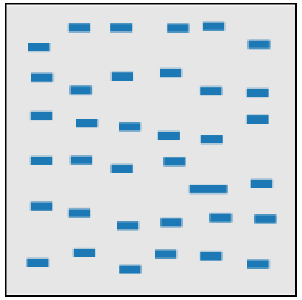
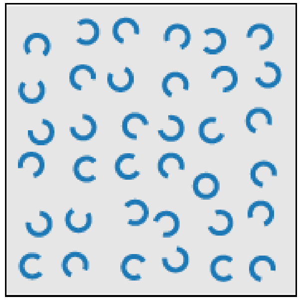
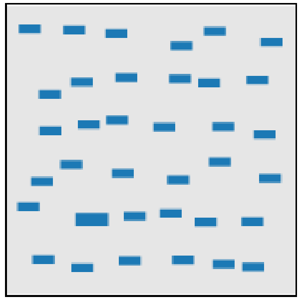
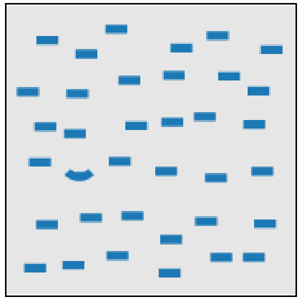
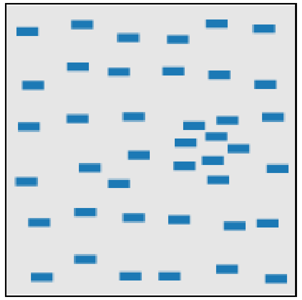
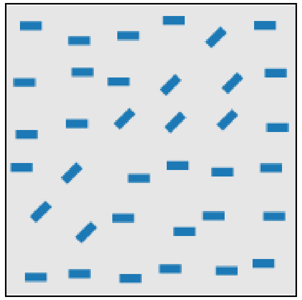
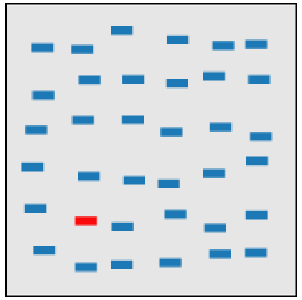
SUBJECT PUNCHED QUICKLY OXIDIZED TCEJBUS DEHCNUP YLKCIUQ DEZIDIXO
CERTAIN QUICKLY PUNCHED METHODS NIATREC YLKCIUQ DEHCNUP SDOHTEM
SCIENCE ENGLISH RECORDS COLUMNS ECNEICS HSILGNE SDROCER SNMULOC

List of Preattentive Visual Properties

See Christopher Healey’s page for examples:

https://www.csc2.ncsu.edu/faculty/healey/PP/

			
line (blob) orientation Julész & Bergen 83; Sagi & Julész 85a, Wolfe et al. 92; Weigle et al. 2000	length, width Sagi & Julész 85b; Treisman & Gormican 88	closure Julész & Bergen 83	size Treisman & Gelade 80; Healey & Enns 98; Healey & Enns 99
			
curvature Treisman & Gormican 88	density, contrast Healey & Enns 98; Healey & Enns 99	number, estimation Sagi & Julész 85b; Healey et al. 93; Trick & Pylyshyn 94	colour (hue) Nagy & Sanchez 90; Nagy et al. 90; D'Zmura 91; Kawai et al. 95; Bauer et al. 96; Healey 96; Bauer et al. 98; Healey & Enns 99

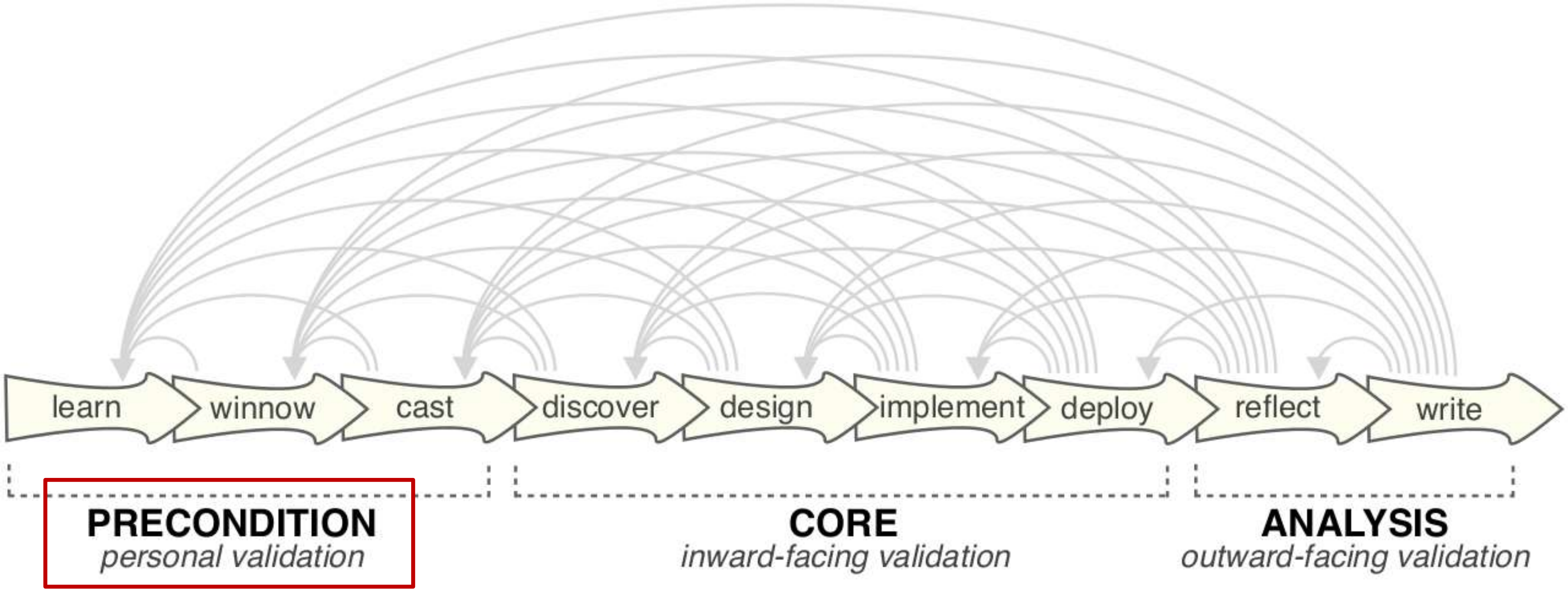
Summary: Marks and Channels

- **Marks** are the geometrical primitives used for visualization
- **Channels** control their appearance
- Channels should be matched to data in order to be **expressive**
- Some channels are more **effective** than others. Use those for the most important attributes!
  – Pathline encoded correlations and expression levels as positions on a common scale, since they were central to the analytic task
- Some channels are processed **pre-attentively**
  – Detected rapidly, independent of distractors
  – Can be used when it is important that certain items will be noticed
  – Combining pre-attentive features makes them lose their “magic”

Section 7.3:
Visualization Design Process

Methodology for Visualization Design

- Nine-stage design methodology [Sedlmair et al. 2012]
  - Guidelines for developing visualization software prototypes for scientific collaborators

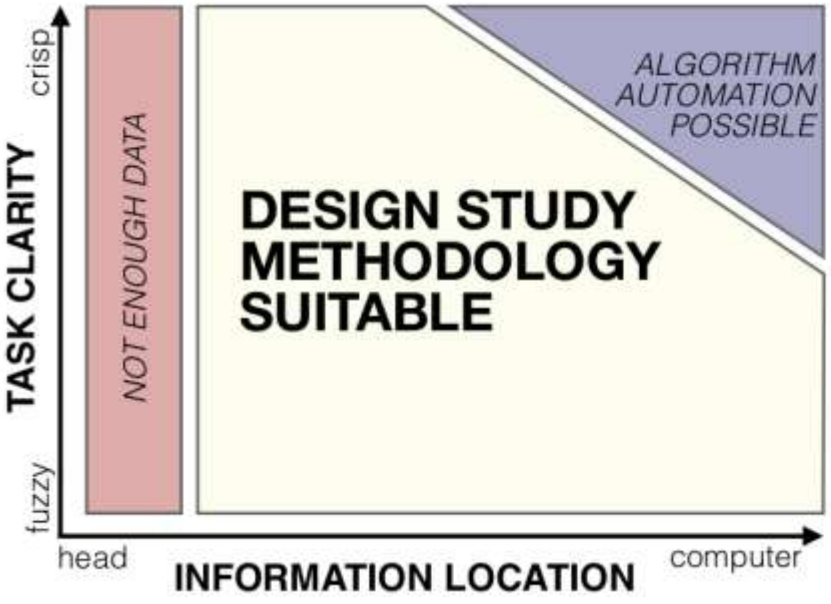


Learn

- **Perception:** Be aware of limitations of the human visual system
- **Techniques:** Have a broad perspective of available tools
  - Explains the sometimes “superficial” treatment of alternative techniques within this lecture
  - Assignments provide opportunity to become familiar with widely applicable software packages
- **Methodology:** Know how to build collaborations and create systems
  - Today’s lecture!
- **Read up** on literature relevant to your project!

Winnow

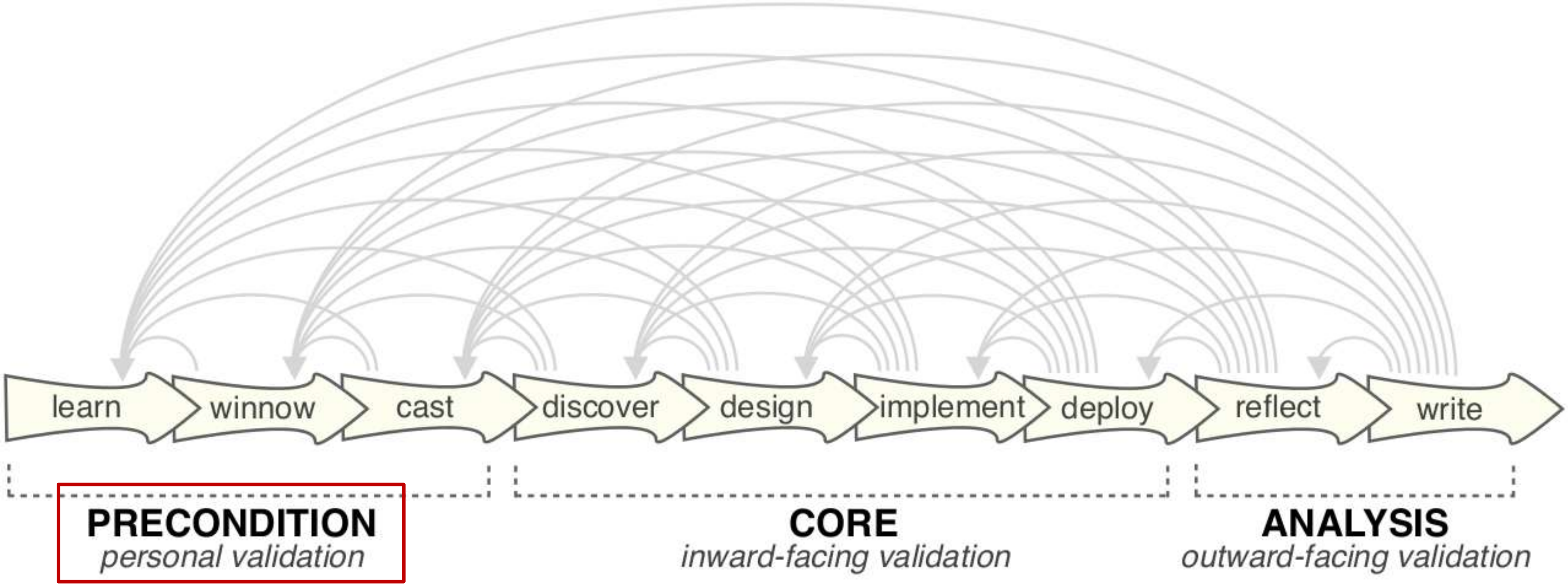
- Selection of collaborators / projects
  - "Talk with many, stay with few"
- **Practical Criteria:**
  - Does data exist, is it enough, can I have it?
  - Can I and can they devote enough time?
- **Intellectual Criteria:**
  - Is the problem interesting and relevant?
  - Do existing solutions suffice?
  - Is visualization suitable?
- **Interpersonal Criteria**



Cast

- Be aware of **important roles** in the project
  - **Front line analyst** (e.g., PhD student)
    - User of your tool
    - Performs day-to-day analysis of the data
    - Often generates the data
  - **Gatekeeper** (e.g., Professor)
    - Grants and denies access to data (and permission to publish about it)
    - Decides on how resources are spent
  - **Fellow tool builders**
    - Avoid relying on their characterization of the problem alone

Core Stage


learn
winnow
cast
discover
design
implement
deploy
reflect
write
PRECONDITION
personal validation
CORE
inward-facing validation
ANALYSIS
outward-facing validation

Discover

- Understand the domain user’s **research questions, data analysis problems, and tasks**
- **Methods:**
  - *Talking:* Iterative (get feedback on your abstraction)
  - *Fly-on-the-wall:* Silently observing
  - *Contextual inquiries:* Interrupt and ask
  - *Reading* domain literature

The 5 Ws of Interactive Visual Data Analysis

[Tominski/Schumann 2020] encourage asking **five W questions:**

- **What** data are to be analyzed?
  – From domain-specific to abstract characterization
- **Why** are the data analyzed?
  – Goals and tasks required to reach them
- **Who** will analyze the data?
  – Account for individual background knowledge, conventions, preferences
- **Where** will the data be analyzed?
  – Classical workstation vs display wall vs head-mounted display vs CAVE
- **When** will the data be analyzed?
  – Integration into workflow of data analyst

Data Abstraction

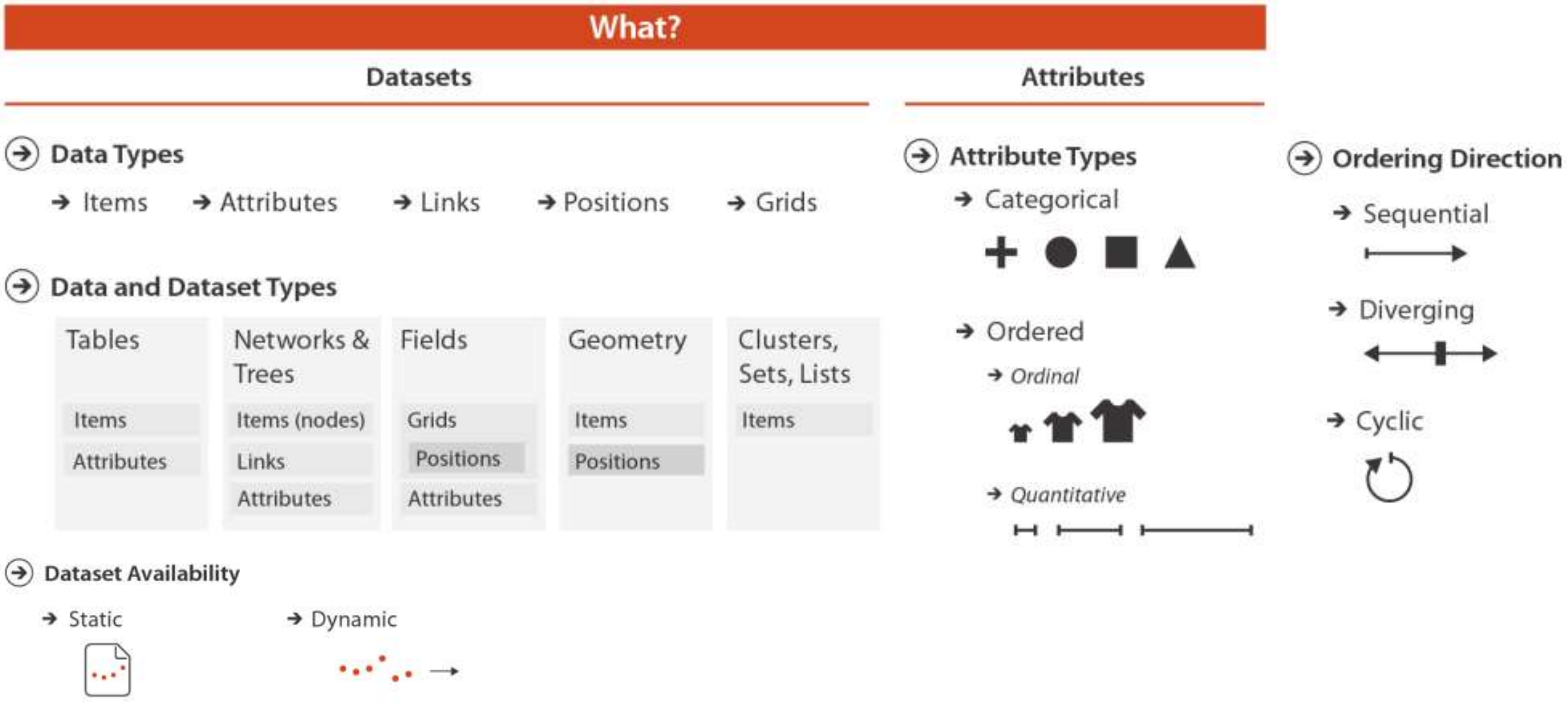


Figure from: Tamara Munzner, Visualization Analysis and Design, 2014

Data Abstraction: Dataset Types

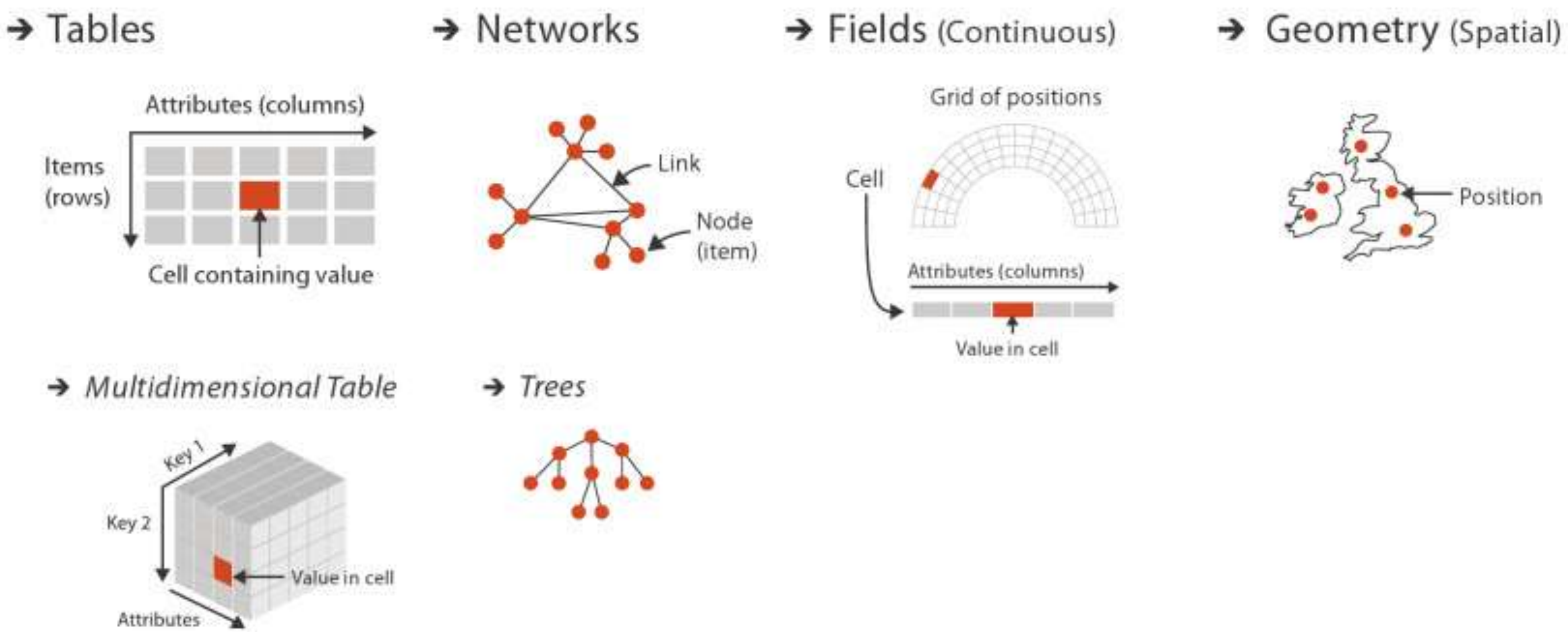
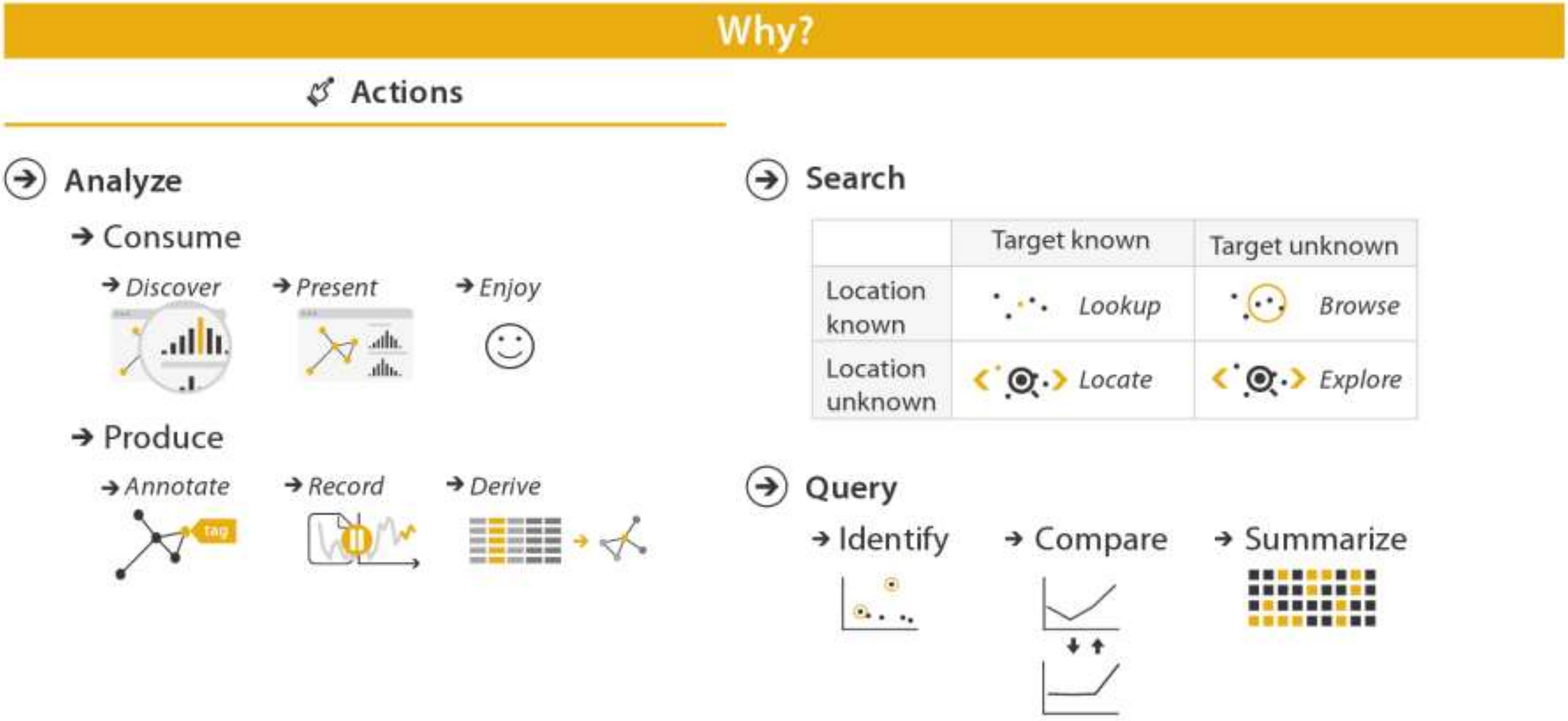


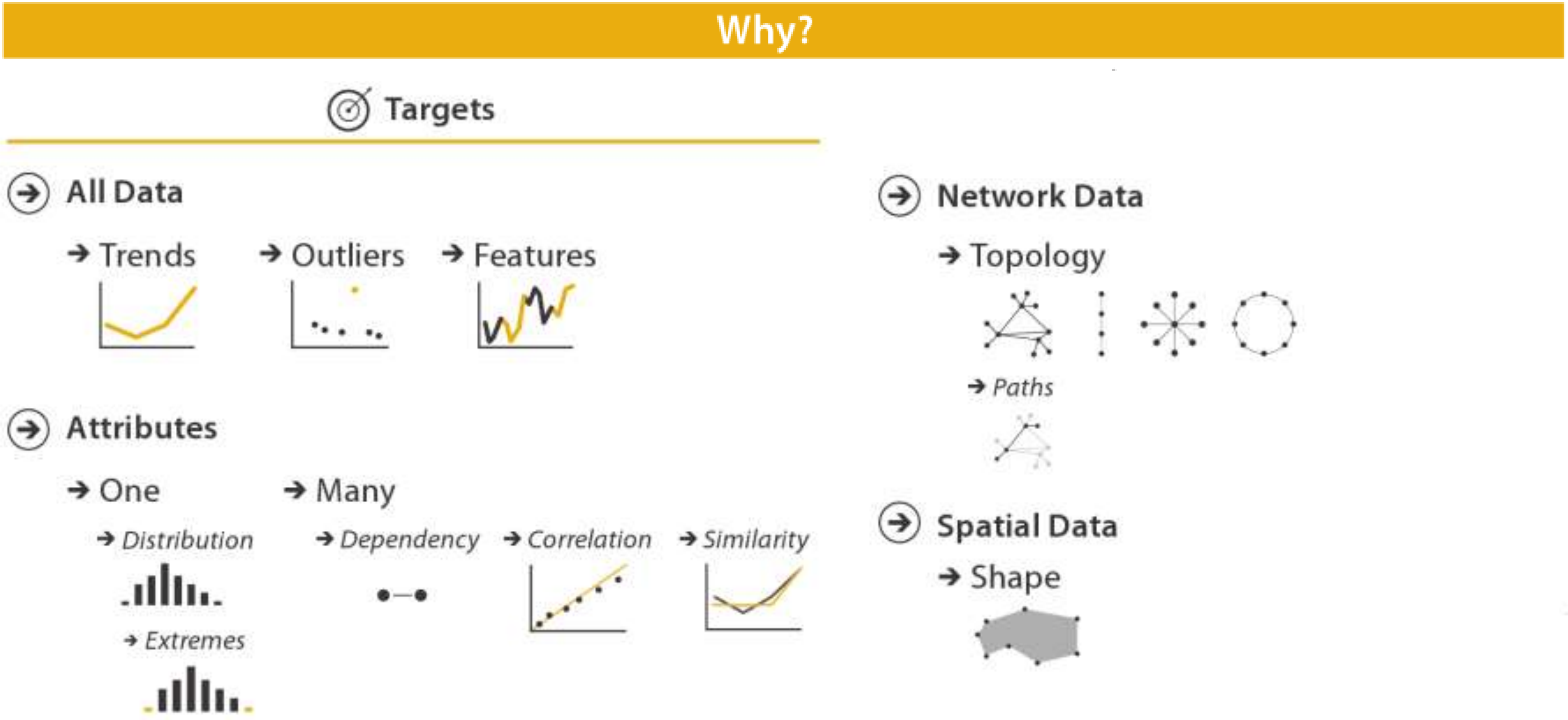
Figure from: Tamara Munzner, Visualization Analysis and Design, 2014

Task Abstraction: Actions



Images from Tamara Munzner, Visualization Analysis and Design, 2014

Task Abstraction: Targets



Images from Tamara Munzner, Visualization Analysis and Design, 2014

Design

- **Goal** of visualization is to create a solution
  - for a specific task
  - performed by a specific group of users
  - on specific data
- **Satisfy** users, do not attempt to **optimize**
  - Contradictory objectives are the norm

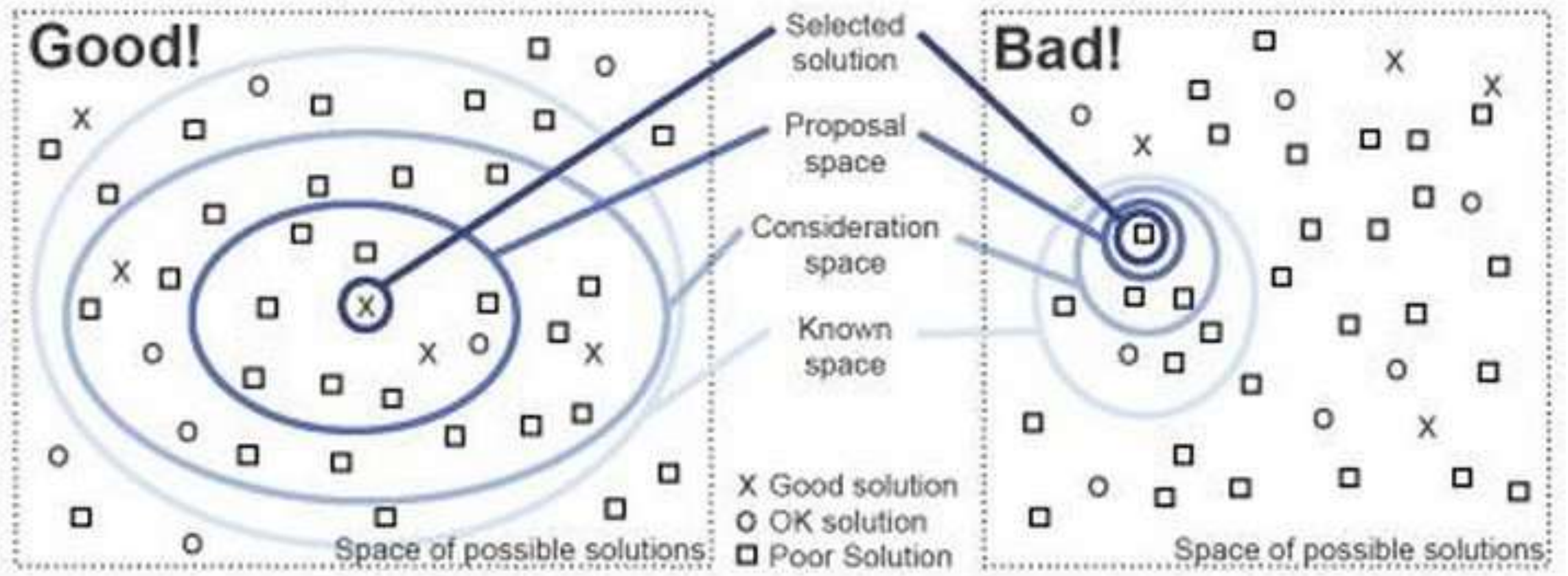


Figure from: Tamara Munzner, Visualization Analysis and Design, 2014

How to Construct a Vis Idiom: Encode

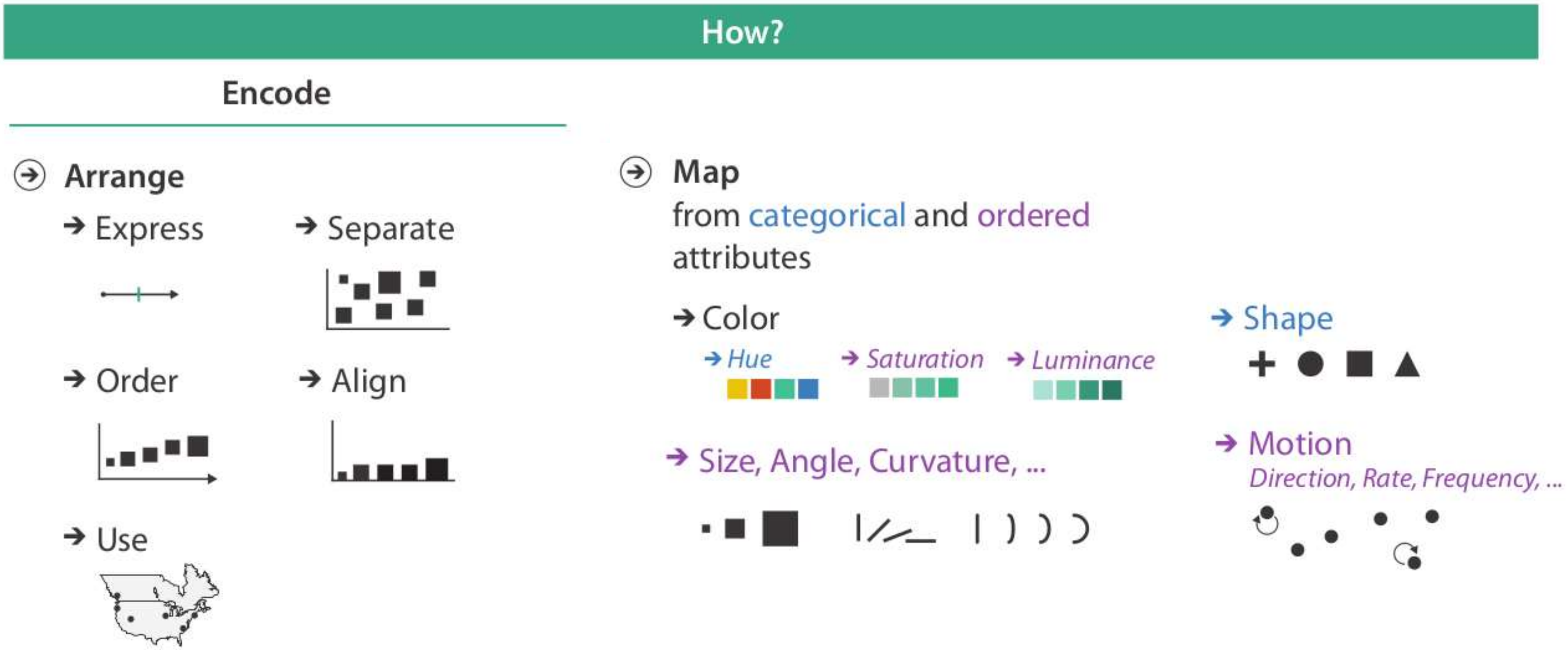


Figure from: Tamara Munzner, Visualization Analysis and Design, 2014

How to Construct a Vis Idiom: Manipulate, Facet, Reduce

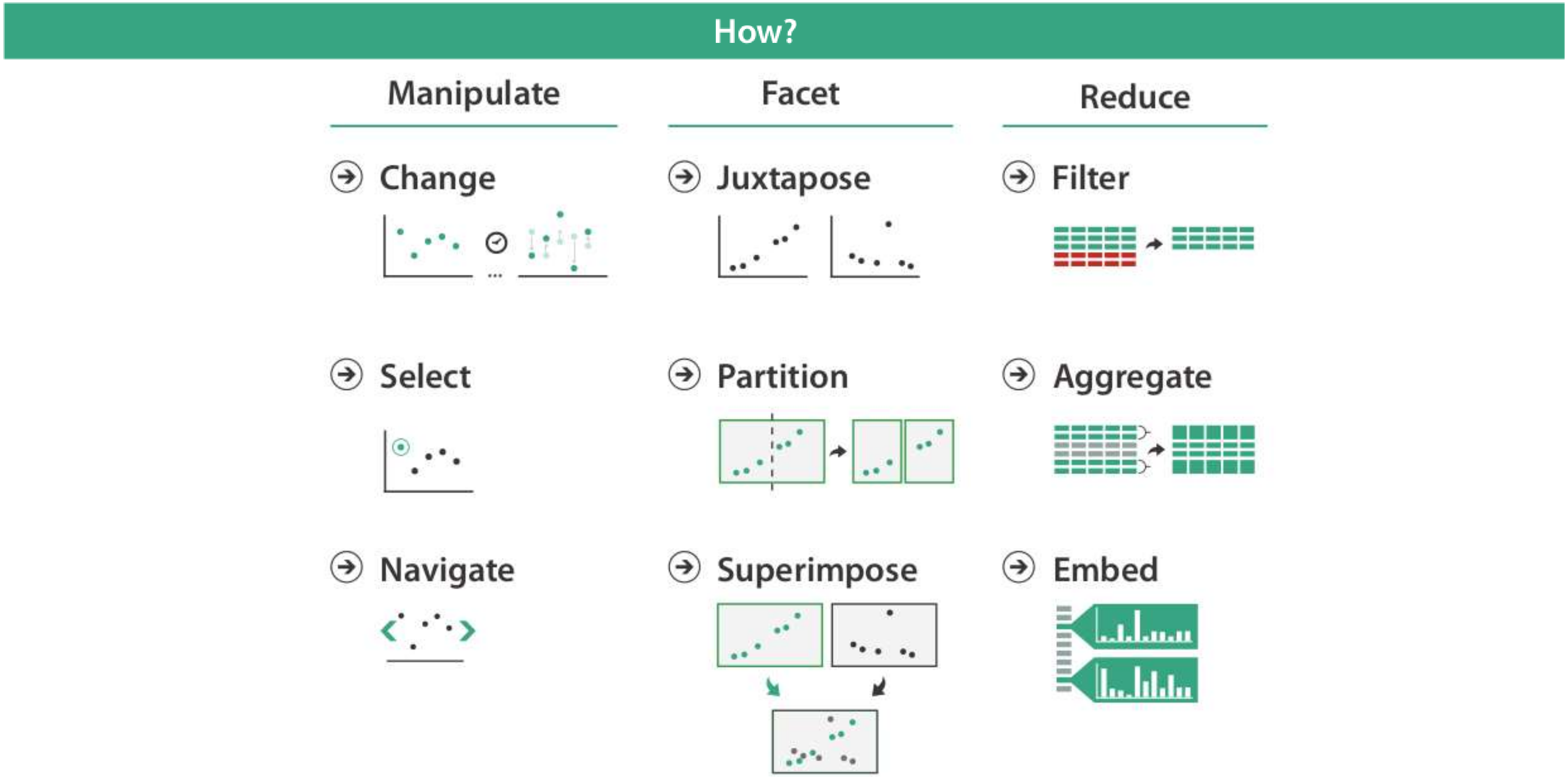


Figure from: Tamara Munzner, Visualization Analysis and Design, 2014

The Shneiderman Mantra

- The design of most systems for information retrieval follows a strategy by [Shneiderman 1996]:
- In addition to **overview**, **zoom**, **filter**, **details-on-demand**, he also lists
  - **Relate**: View and follow relationships between items
  - **History**: To support undo, replay, refine
  - **Extract**: Export the selected information to a file, or share with others

> There are many visual design guidelines but the basic principle might be summarized as the Visual Information Seeking Mantra:
>
> Overview first, zoom and filter, then details-on-demand
> Overview first, zoom and filter, then details-on-demand
> Overview first, zoom and filter, then details-on-demand
> Overview first, zoom and filter, then details-on-demand
> Overview first, zoom and filter, then details-on-demand
> Overview first, zoom and filter, then details-on-demand
> Overview first, zoom and filter, then details-on-demand
> Overview first, zoom and filter, then details-on-demand
> Overview first, zoom and filter, then details-on-demand
> Overview first, zoom and filter, then details-on-demand

Implement

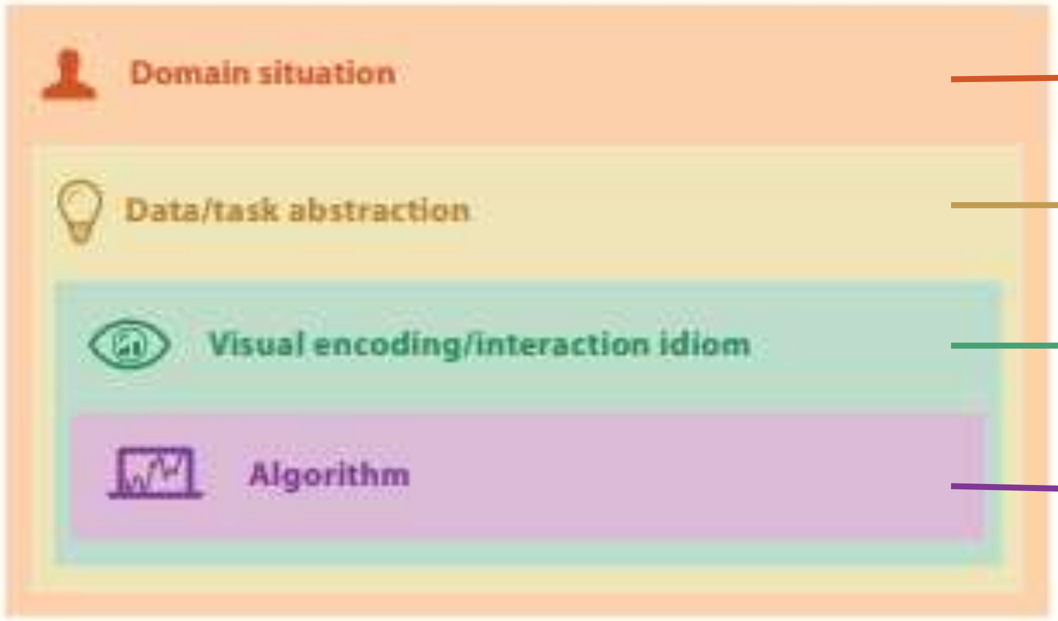
- **Prototypes** help confirm that you’ve correctly analyzed the users’ needs
  - Avoid investing lots of time in creating a software that ends up not being used
  - Start with paper prototypes / “mock-ups”
  - Write “throw-away” code simply and rapidly
  - Make use of existing infrastructure
- Find middle ground on **usability**
  - With too little usability, tool will not be used
  - Usability should not push utility out of focus

Deploy

- Release your software "into the wild"
- **Validate** benefit from tool:
  - Faster or more accurate analysis
  - New insights (which might enable a higher degree of automation)
- **Case study:** Solve specific real problem with real users and real data
  - **Usage scenario:** Real data, analysis done by developer of tool
  - **Pair analytics:** Insight gained in tandem by computer / data and domain scientist

Four Nested Level Model of Vis Design

- [Munzner 2014] distinguishes **four levels** of visualization design
  - They are *nested*: even if inner ones are solved perfectly, the overall system will not be used if we made incorrect choices at an outer level
  - Overall design process remains iterative despite this



Threat: You misunderstood their needs

Threat: You're showing them the wrong thing

Threat: The way you show it doesn't work

Threat: Your code is too slow

Figure from: Tamara Munzner, Visualization Analysis and Design, 2014

Matching Validation to Contribution

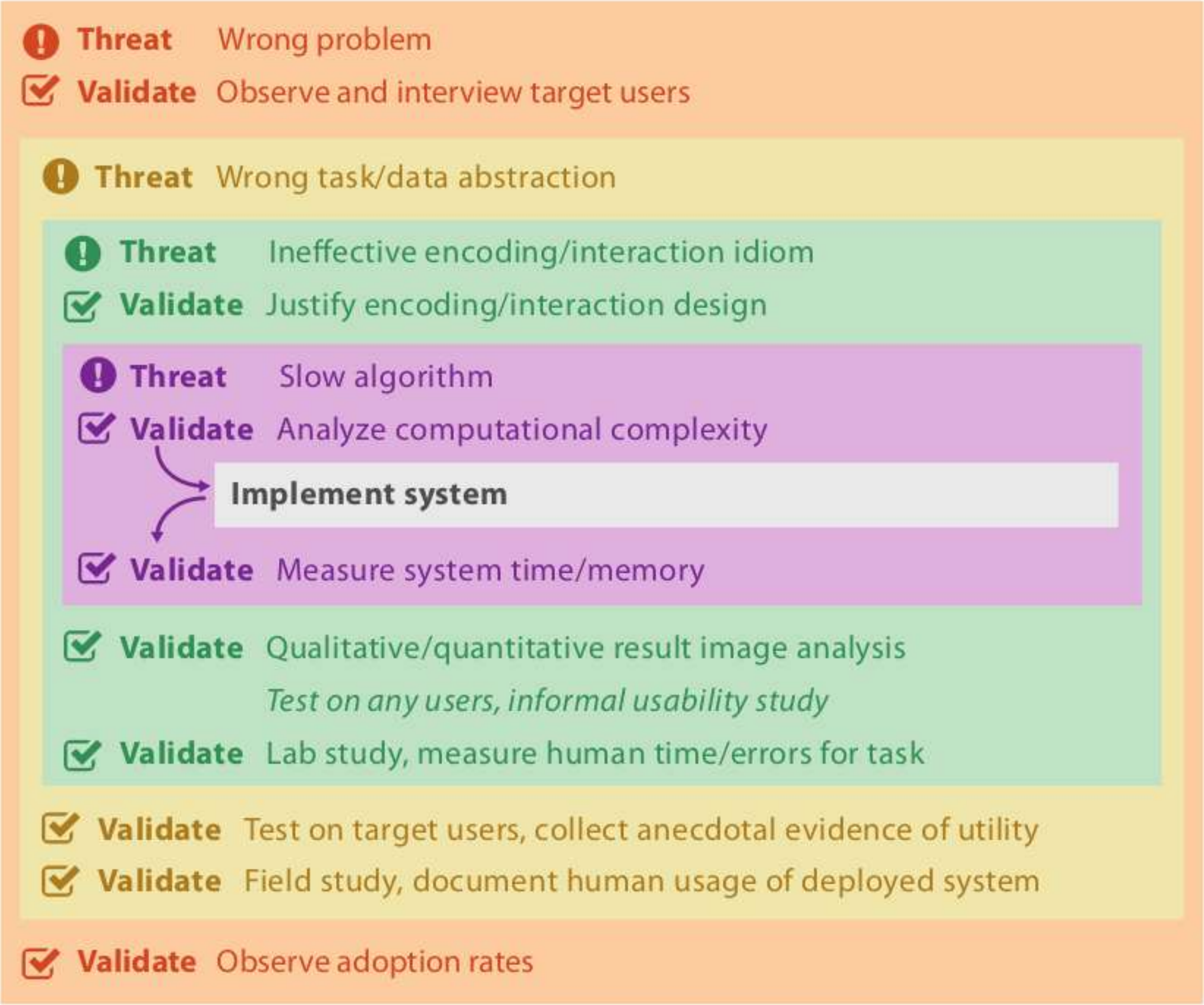
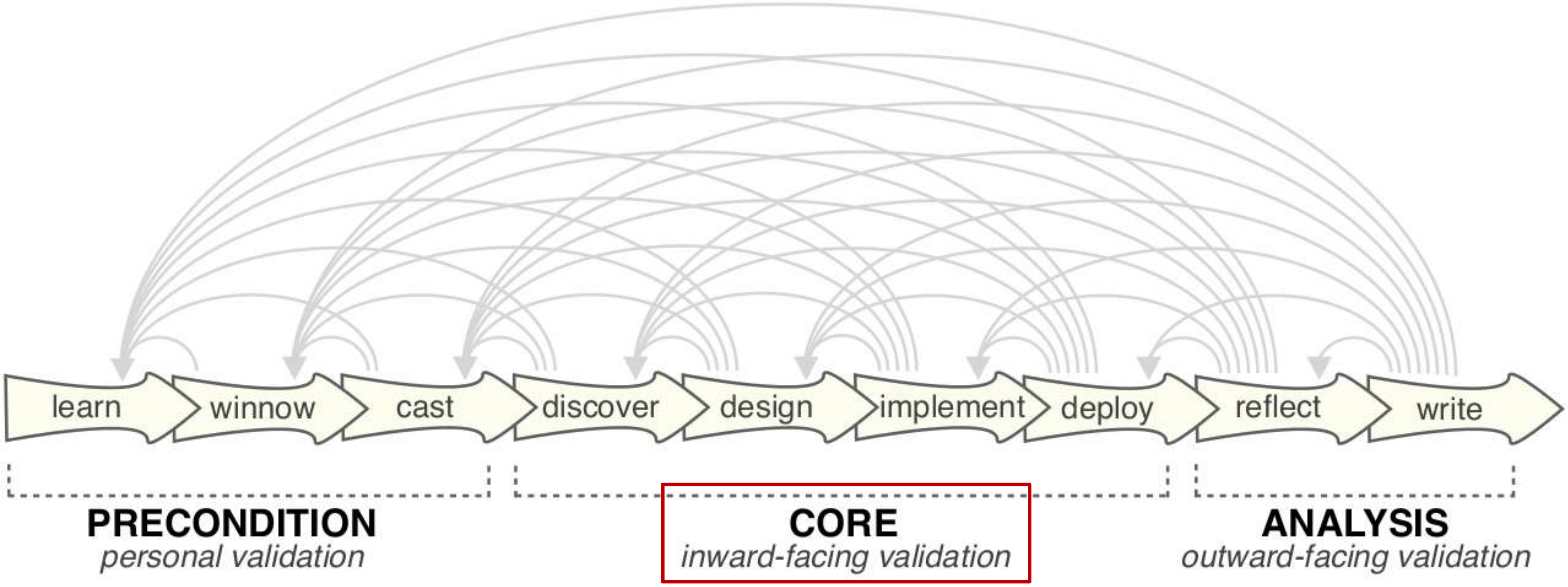


Figure from: Tamara Munzner, Visualization Analysis and Design, 2014

- Most individual visualization research projects only contribute something novel at one or a few levels
  - Rely on existing techniques or established task abstraction for the others
- Type of validation needs to match the level of contribution
  - Can be *immediate* or *downstream*

Analysis Stage

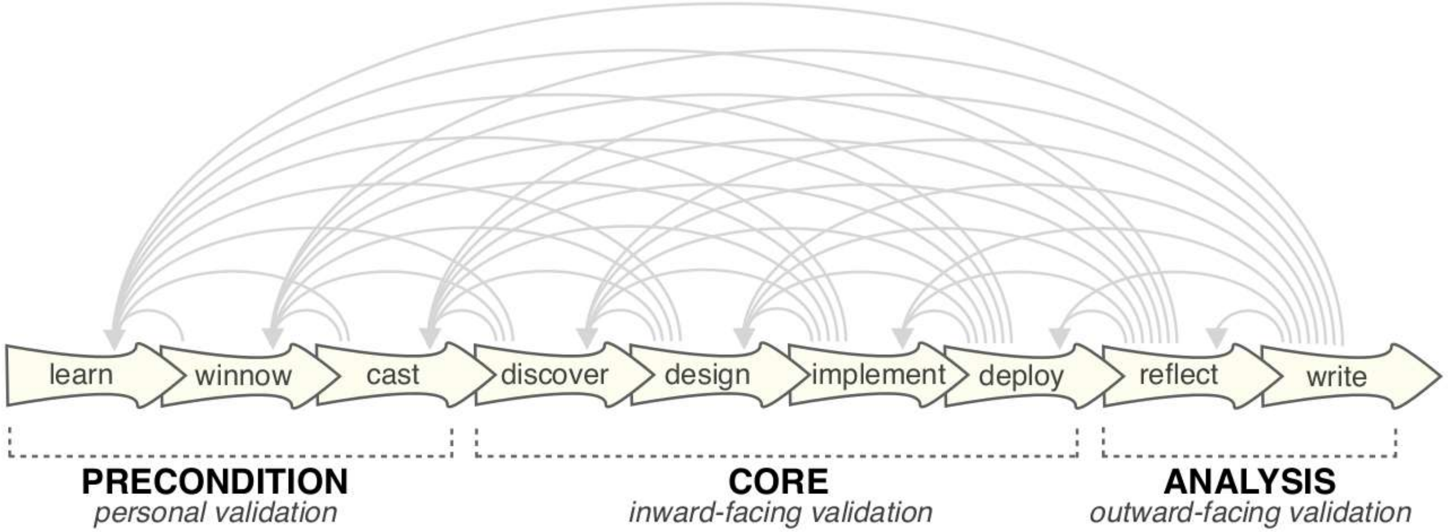


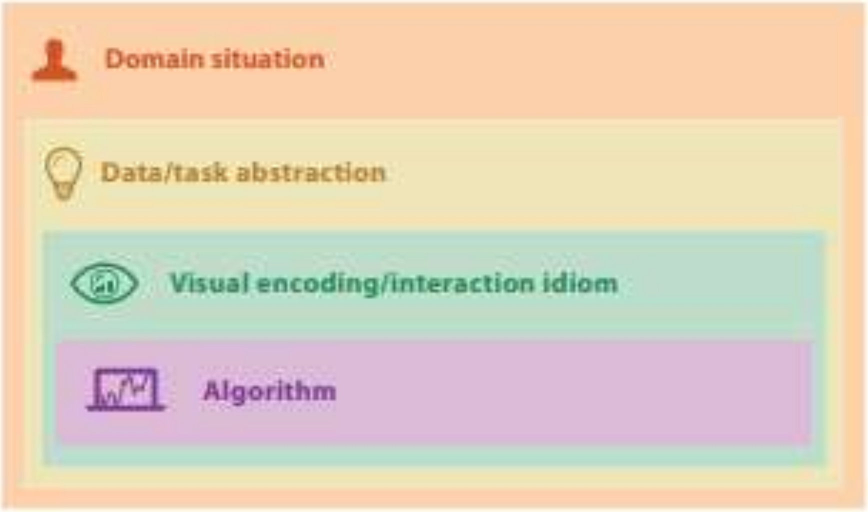
Reflect / Write

- **Reflect** on what you’ve learned from building the tool
  - How to do better next time?
  - Confirm, refine, reject, propose guidelines
- In an academic context, **report** on your tool in a scientific paper
  - Describe the tool and evaluation
  - Justify design choices
  - Discuss lessons learned

Summary: Visualization Design Process

- **Nine-stage design methodology** provides useful guidance / structure for interdisciplinary collaborations
  - Helps avoid missing important steps
- **Four nested level model** helps match validation to contribution

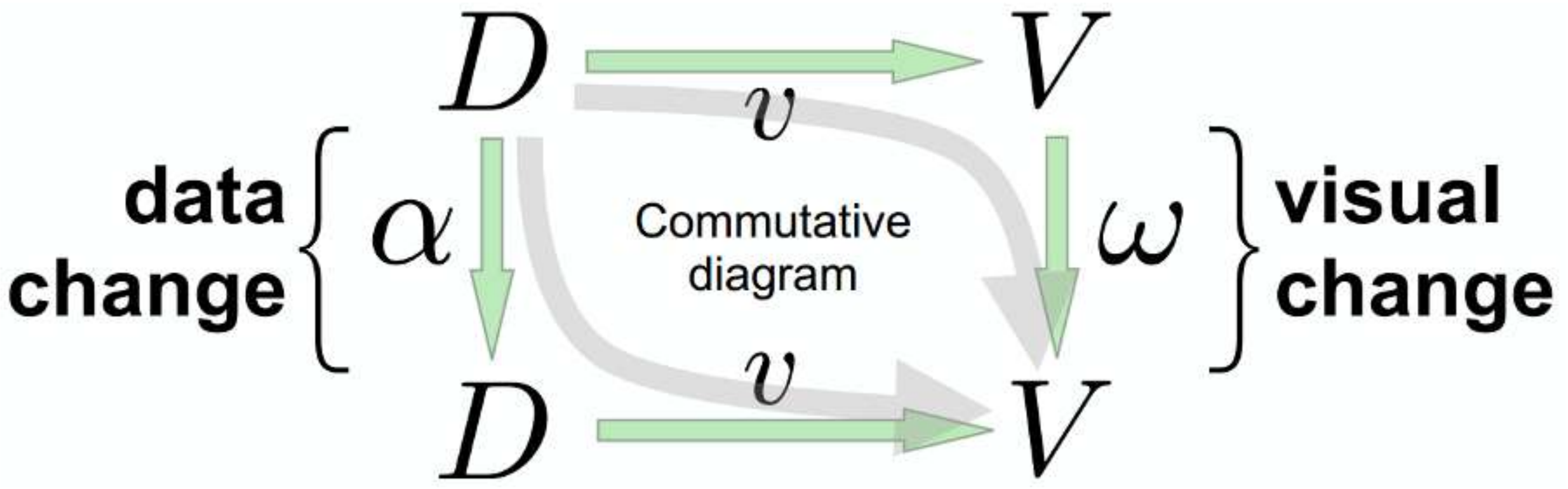




Section 7.4:
An Algebraic Perspective

Algebraic Perspective on Visualization Design

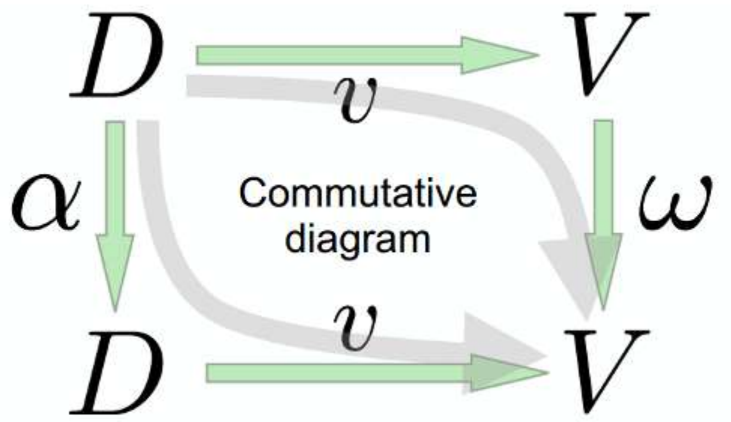
- Most guidelines focus on data types and tools
- **Alternative** [Kindlmann/Scheidegger 2014]: Consider correspondence between changes in data and image
  - Tool for immediate validation of visual encodings



Images from Gordon Kindlmann

Derived Design Principles

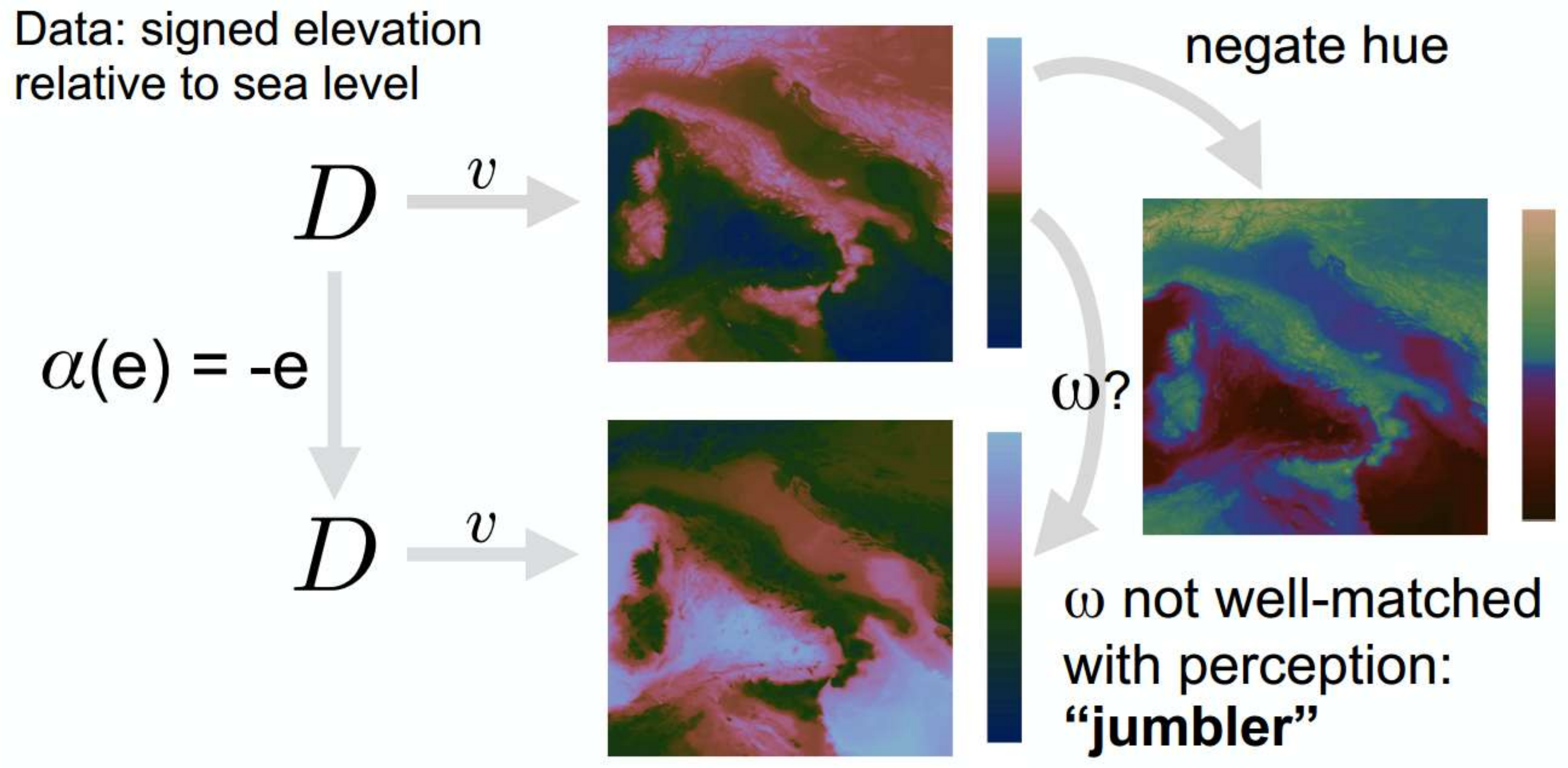
- **Basic question:**
  „Are α and ω well-matched?"



1. Principle of Visual-Data Correspondence
   – Does ω make sense, given α?
2. Principle of Unambiguous Data Depiction
   – For all important α, is ω obvious?
3. Principle of Representative Invariance
   – Can obvious ω arise without data change (α=1)?

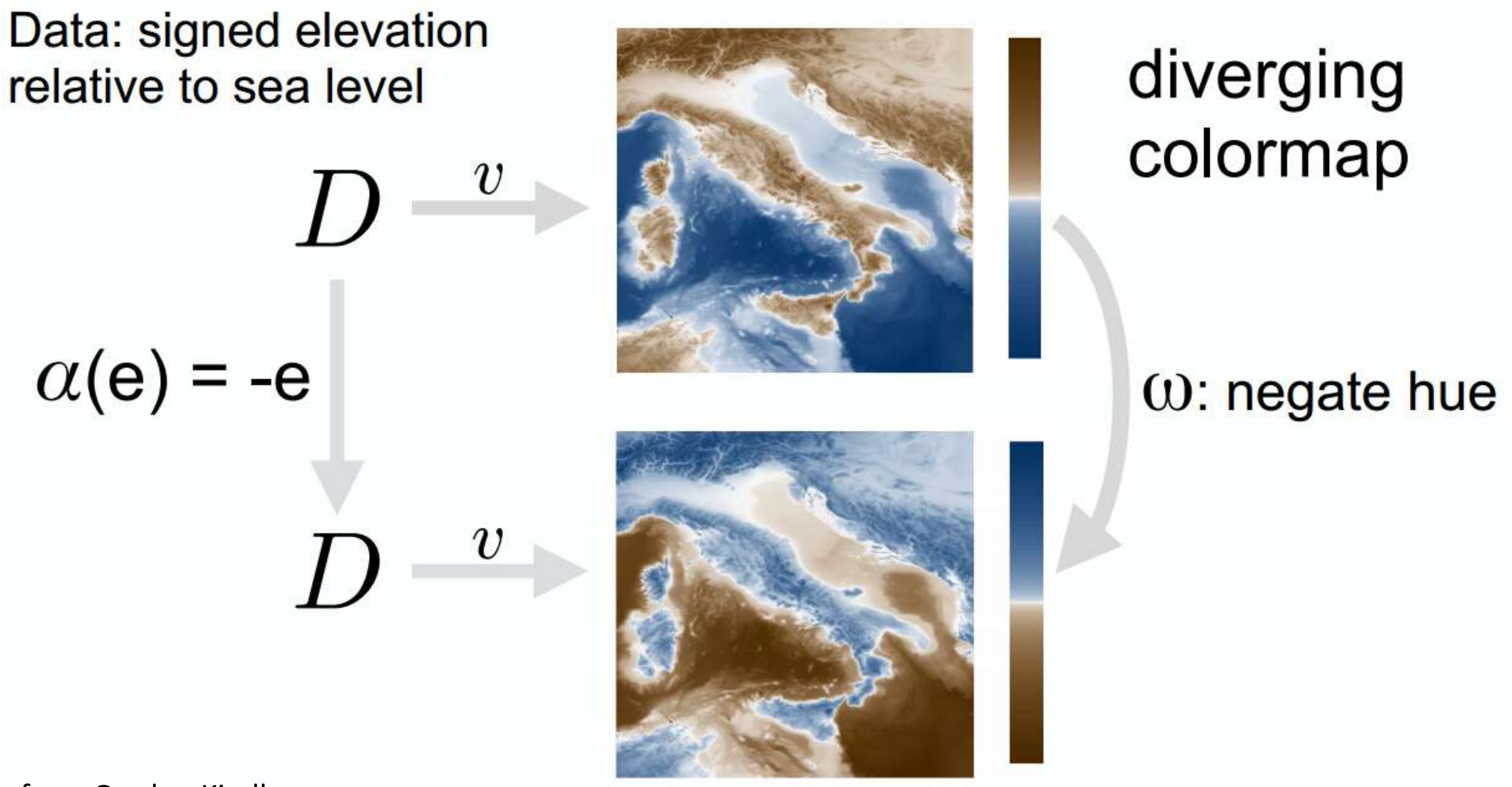
Images from Gordon Kindlmann

Correspondence: Elevation Colormap



Images from Gordon Kindlmann

Correspondence: Diverging Colormap



Images from Gordon Kindlmann

Correspondence: Scatter Plots

Data: % men vs women employed as senior managers in various countries

$D \xrightarrow{v}$

α: decrease gender gap for one country: EST

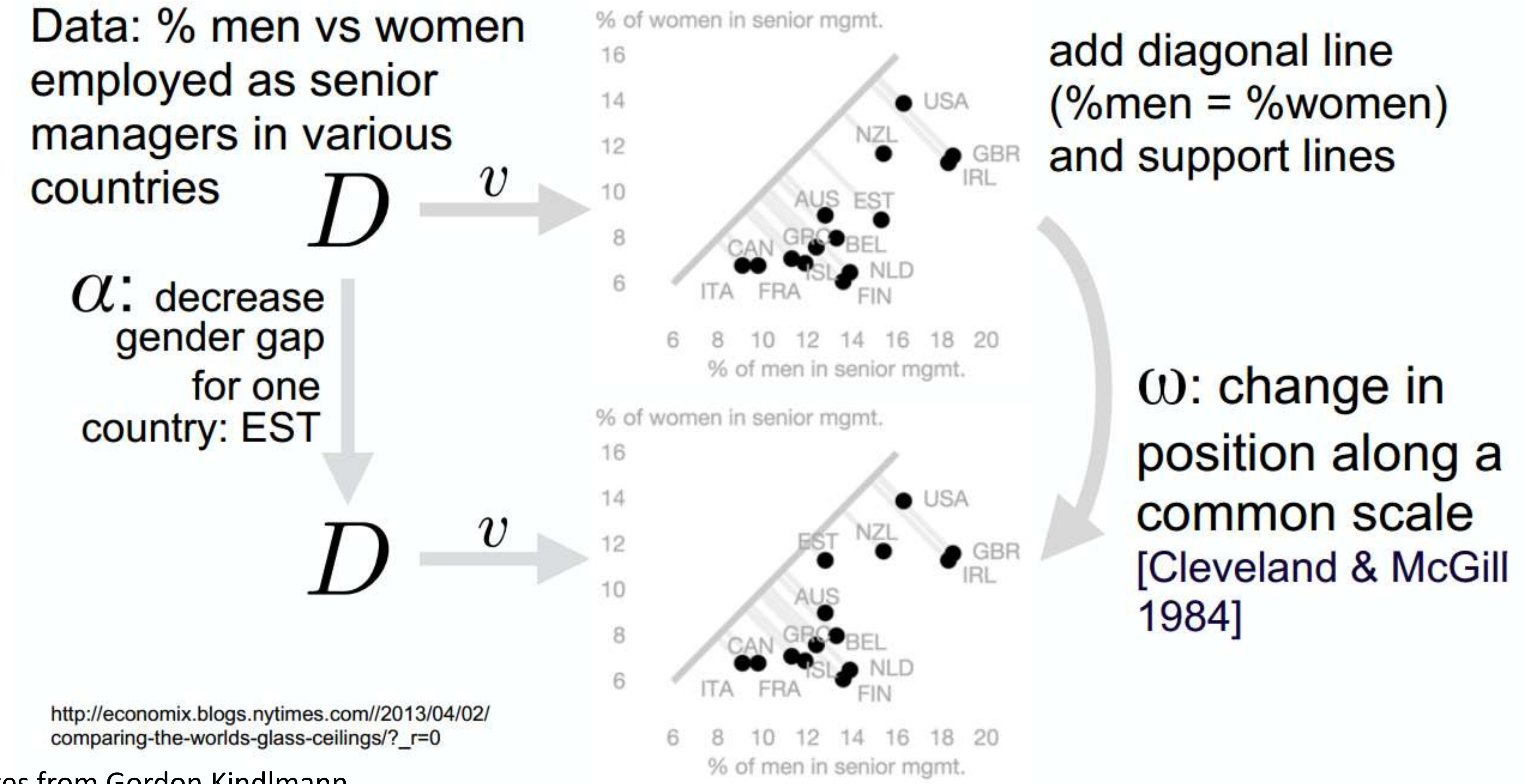
$D \xrightarrow{v}$

ω? Not clear how big that change was

http://economix.blogs.nytimes.com//2013/04/02/comparing-the-worlds-glass-ceilings/?_r=0

Images from Gordon Kindlmann

Correspondence: S-Plots with Support Lines



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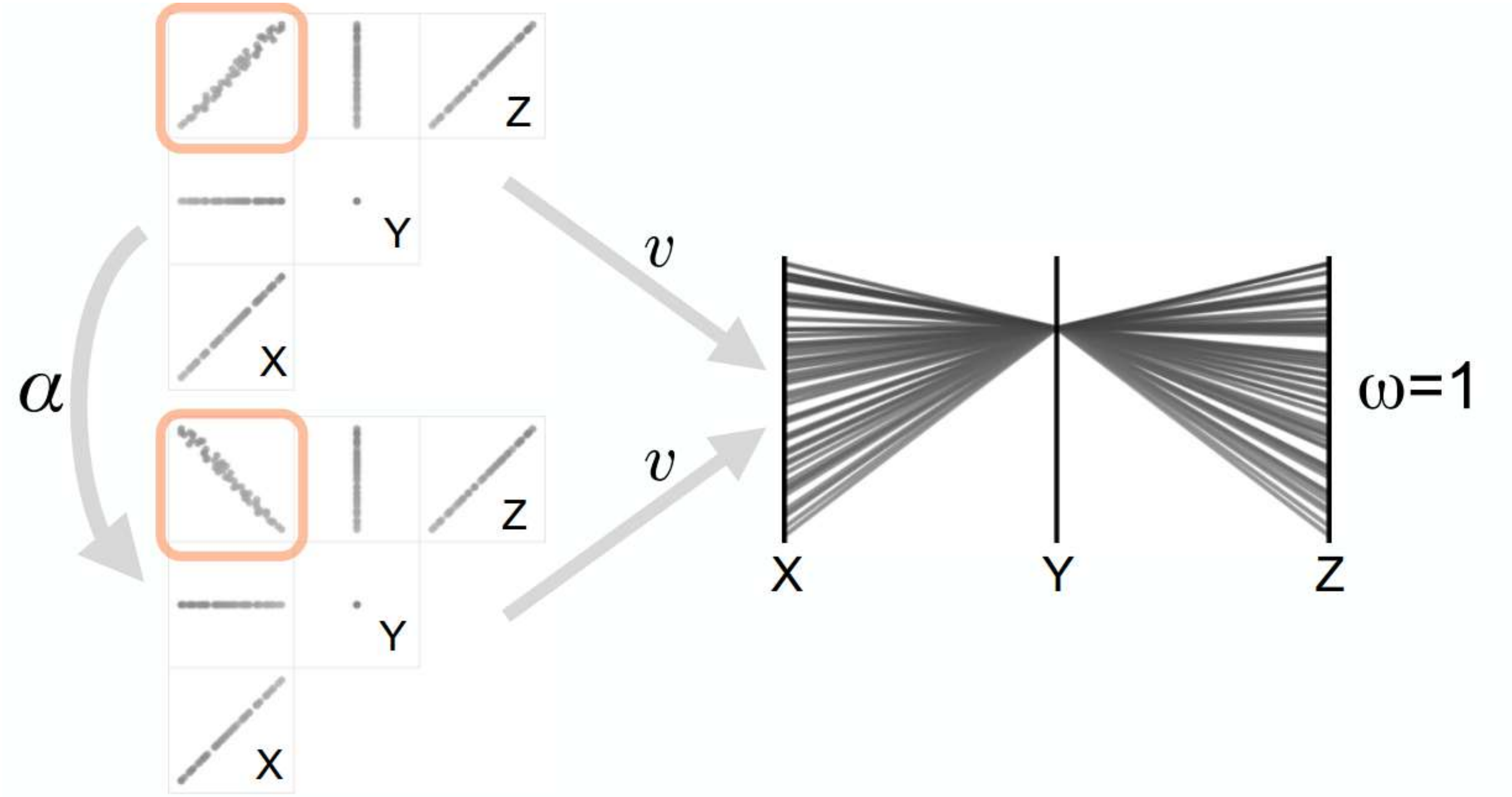
Unambiguity: Parallel Coordinates



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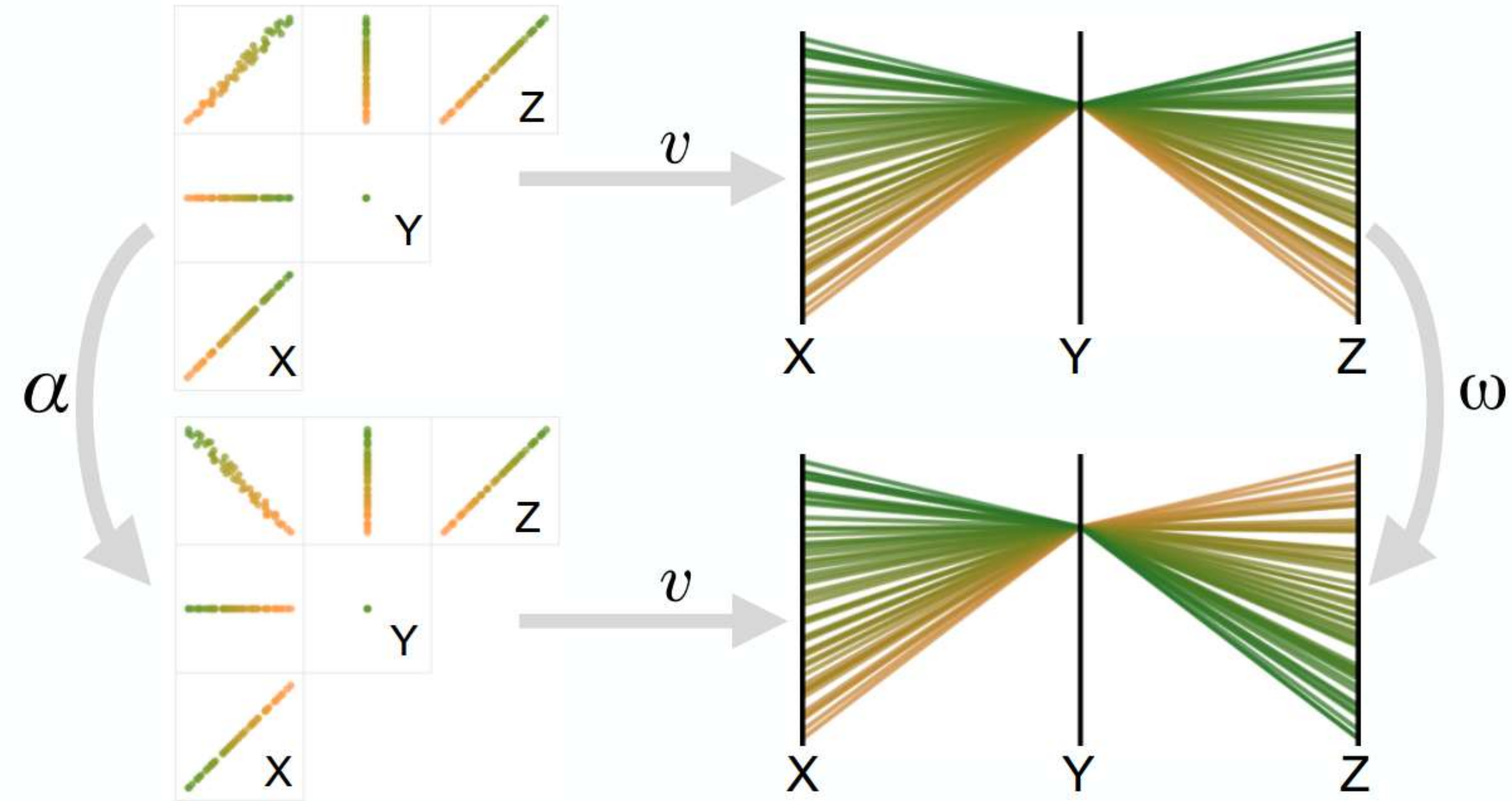
Unambiguity: Parallel Coordinates with Color



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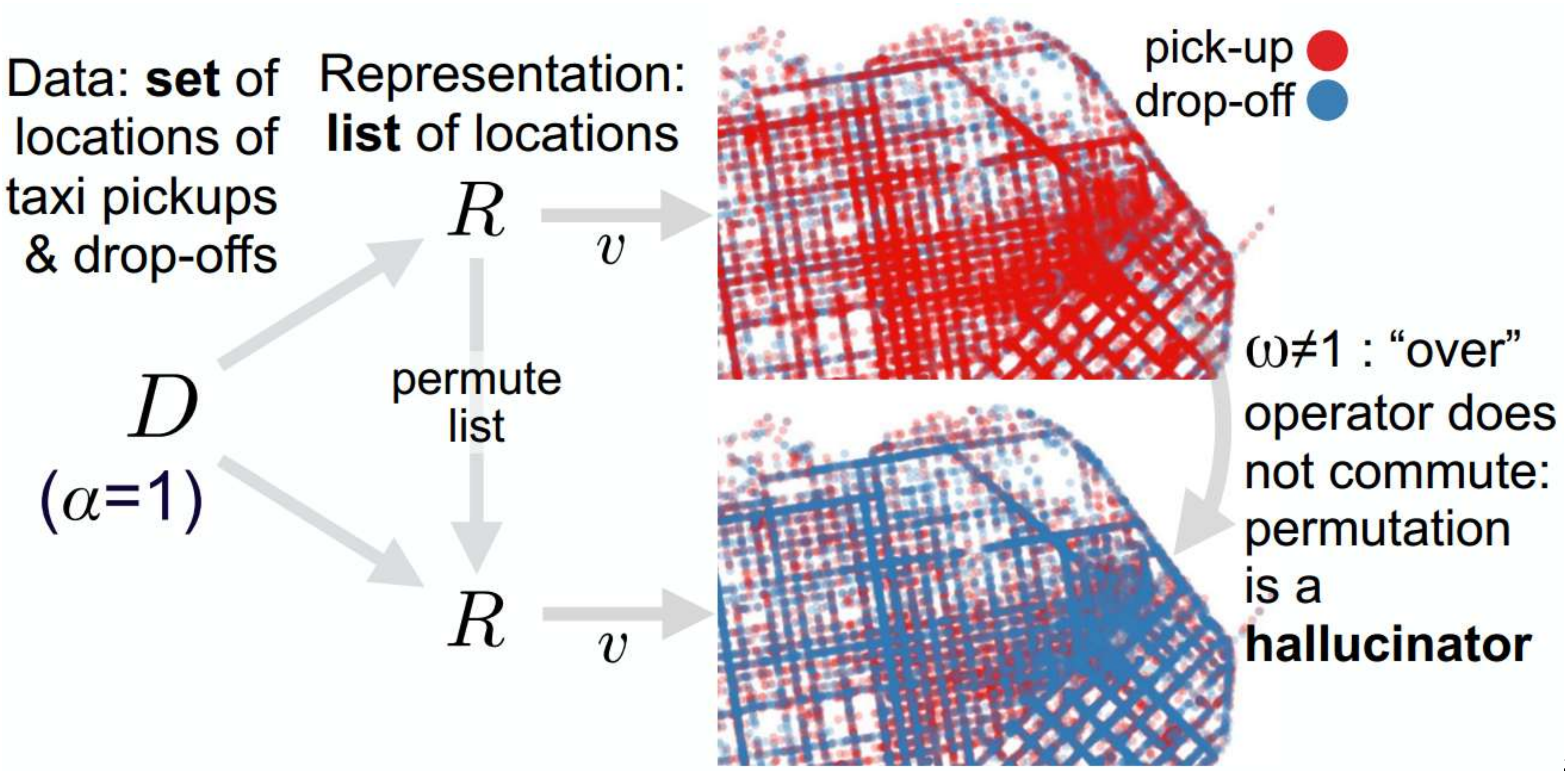
Invariance: Alpha-blended Marks



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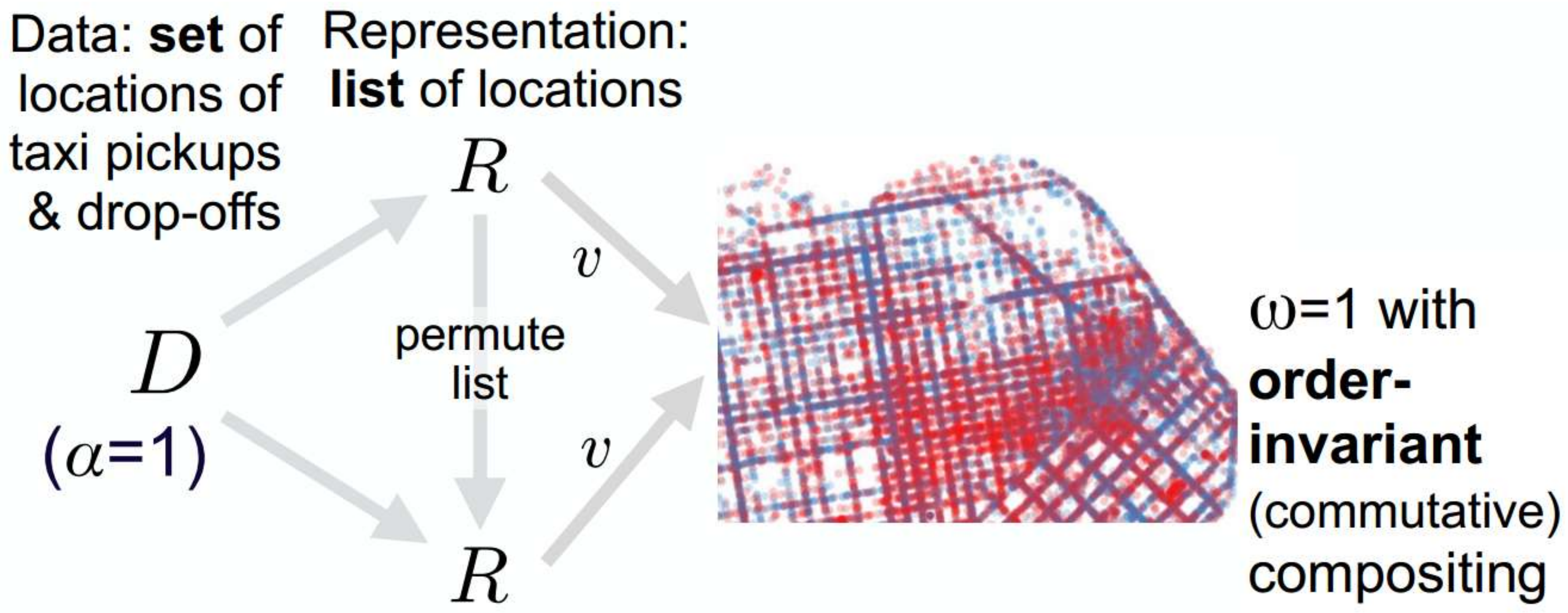
Invariance: Commutative Compositing



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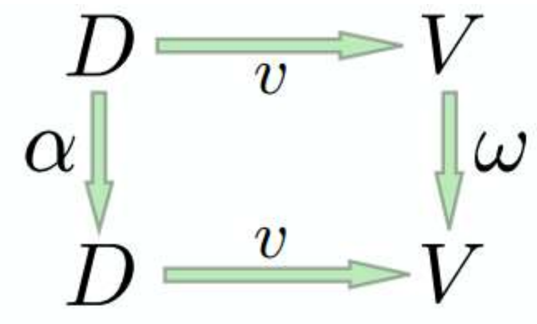
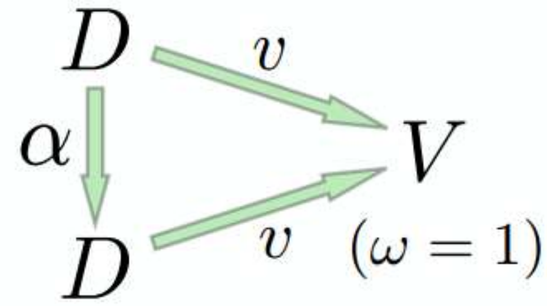
Summary: Algebraic Vis Design

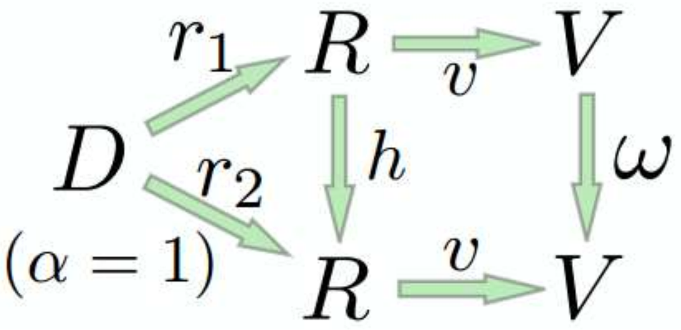
- Make sure your visualization design
  - **has meaningful visual-data correspondence** (important changes should not be *jumbled*; visualization should not *mislead* regarding importance of small changes)



  - **depicts the data unambiguously** (different data should not be *confused*)



  - **is invariant to representation changes** (for different representations of the same data, no changes should be *hallucinated*)



Images from Gordon Kindlmann

References

- Meyer et al., **Pathline: A Tool For Comparative Functional Genomics,** Computer Graphics Forum 29(3), 2010
- Tamara Munzner, **Visualization Analysis and Design,** A K Peters, 2014
- Sedlmair et al., **Design Study Methodology: Reflections from the Trenches and the Stacks,** IEEE Trans. Vis. Comp. Graph. 18(12):2431-2440, 2012
- Kindlmann/Scheidegger, **An Algebraic Process for Visualization Design,** IEEE Trans. Vis. Comp. Graph. 20(12):2181-2190, 2014

Exam-Like Questions

1. Name a magnitude channel (for ordered attributes) and an identity channel (for categorical attributes)
2. Is position on a common scale or 2D area more effective as a magnitude channel? Why?
3. Name a type of threat in visualization design, and a validation method that can guard against it.
4. Describe a visualization that violates the principle of unambiguous data depiction, and suggest a way to improve it.